ALCOHOLS

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Alcohols

Basic Information about Alcohol Group:

Classification:

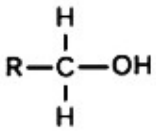
Alcohols are classified into two ways.

(a) Degree of an alcohol : Nature of alcohol based on –OH group attached to the type of carbon.

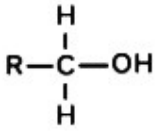
- **(i)** 1° alcohol : Those alcohols in which –OH group is attached to 1° carbon atom.
- **(ii)** 2° alcohol : Those alcohols in which –OH group is attached to 2° carbon atom.
- **(iii)** 3° alcohol : Those alcohols in which –OH group is attached to 3° carbon atom.

Introduction

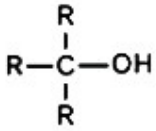
An alcohol is a compound that contains one or more hydroxyl groups having general formula ROH or $C_nH_{2n+1}OH$ where R is an alkyl or substituted alkyl group. The group can be primary, secondary or tertiary. It may be open chain or cyclic; it may contain a double bond, an halogen atom or an aromatic ring.



1° alcohol



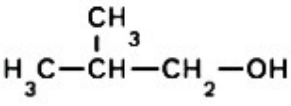
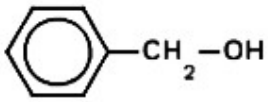
2° alcohol



3° alcohol

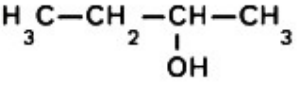
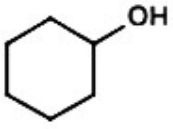
(b) Nature of alcohol based on number of –OH groups present in an alcohol molecule.

- **(i) Monohydric alcohols:** Those in which only on –OH group is present in their molecule are known as monohydric alcohols
- **(ii) Dihydric alcohols :** Those alcohols in which two –OH groups are present in their molecule are known as dihydric alcohols
- **(iii) Trihydric alcohols :** Those alcohols in which three –OH groups are present in their molecule are known as trihydric alcohols
- **(iv) Polyhydric alcohols :** Those alcohols in which more than three –OH groups are present in their molecule are known as polyhydric alcohols.

Example Primary alcohol  and 

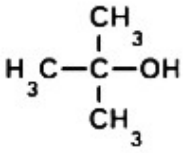
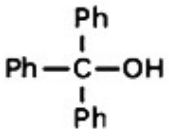
2 – Methyl – 1 – propanol

Benzyl alcohol

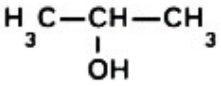
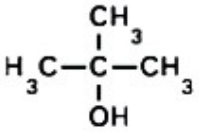
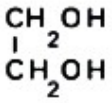
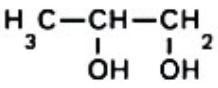
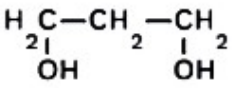
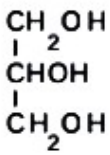
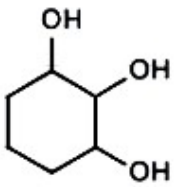
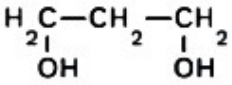
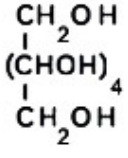
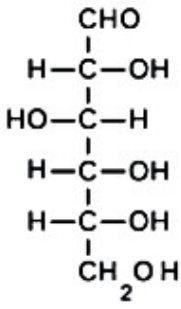
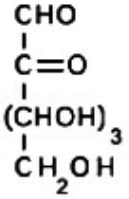
Example Secondary alcohol  and 

2 – Butanol

Cyclohexanol

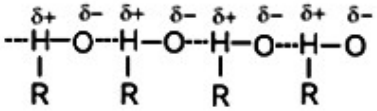
Example Tertiary alcohol  and 

2 – Methyl – 2 – propanol Triphenyl methanol

Alcohol	(i)	(ii)	(iii)
Monohydric alcohol	CH_3CH_2OH Ethyl alcohol	 Propan–2–ol	 2–Methylpropan–2–ol
Dihydric alcohol	 (Ethane–1,2–diol)	 Propylene glycol (Propane–1,2–diol)	 Trimethyleneglycol (Propane–1,3–diol)
Trihydric alcohol	 Gycerol (Propane–1,2,3–triol)	 Cyclohexane–1,2,3–triol	 Trimethyleneglycol (Propane–1,3–diol)
Polyhydric alcohol	 Sorbital	 Glucose	 Fructose

Physical Properties of Alcohols

1. **Physical State:** At ordinary temperature lower alcohols are colourless liquids with distinct smell but the higher homologues are colourless, odourless waxy solids.
2. **Boiling Points:** The boiling points of alcohols are much higher than the corresponding aliphatic hydrocarbons and alkyl halides. Alcohols can undergo intermolecular hydrogen bonding. As a result, alcohols exit as associated molecules rather than discret molecules as shown below



Hence a large amount of energy is required to break these hydrogen bonds. As a result, their boiling points are much higher than the corresponding hydrocarbons, ethers and alkyl halides which do not form hydrogen bonds. Amongst isomeric alcohols, boiling points decrease in the order : 1° > 2° > 3°. However with increase in molecular weight, the boiling points increase regularly.

3. **Solubility :**

The lower alcohols are highly soluble in water but with increase in molecular weight the solubility decrease. Amongst isomeric alcohols, with increase in branching the solubility increases. So, the solubility order of Butanols in t-Butyl alcohol > s-butyl alcohol > Isobutyl alcohol > n-butyl alcohol.

Alcohols and Acids:

We know that an alcohol can accept hydrogen ion to form the protonated alcohol, $\overset{\oplus}{ROH_2}$, thus acts as a base. But when it acts as an acid, loses a hydrogen ion to form the alkoxide ion, RO^-. Since an alcohol contains hydrogen bonded to the very electronegative element oxygen, we would expect it to show appreciable acidity. The polarity of the O–H bond should facilitate the separation of the relatively positive hydrogen as the ion; viewed differently, electronegativity oxygen should readily accommodate the negative charge of the electrons left behind.

The acidity of alcohols is shown by their following reactions.

Example $\quad R-OH + Na \longrightarrow R-\overset{\ominus}{O}\overset{\oplus}{Na} + \frac{1}{2}H_2$

Example $\quad R-OH + R'MgX \longrightarrow R'-H + Mg(OR)X$

Stronger acid Weaker acid

With the possible exception of methanol, they are weaker acids than water, but stronger acids than acetylene or ammonia.

$$R-\overset{\ominus\,\oplus}{O}Na + H-OH \longrightarrow Na^{+} + OH^{-} + R-OH$$

Stronger Base	Stronger Acid	Weaker Base	Weaker Acid

Relative acidities : $H_2O > ROH > HC \equiv CH > NH_3 > RH$

Relative basicities : $\overset{\ominus}{O}H < \overset{\ominus}{O}R < HC \equiv \overset{\ominus}{C} < \overset{\ominus}{N}H_2 < \overset{\ominus}{R}$

Not only does the alkyl group make an alcohol less acidic than water, but more the number of alkyl group, the less acidic the alcohol; methanol is the stronger acid and tertiary alcohol are the weakest.

Distinction between primary, secondary and tertiary Alcohols(Lucas test):

Whether an alcohol is primary, secondary or tertiary is shown by the Lucas test, which is based upon the difference in reactivity of the three classes towards hydrogen halides.

Alcohols (of not more than six carbons) are soluble in the Lucas reagent, a mixture of concentrated hydrochloric acid and zinc chloride. The corresponding alkyl chorides are insoluble. Formation of a chloride from an alcohol is indicated by the cloudiness that appears when the chloride separates from the solution; hence, the time required from cloudiness to appear is a measure of the reactivity of the alcohol.

Observations

A tertiary alcohol reacts immediately with Lucas reagent, and a secondary alcohol reacts within five minutes; a primary alcohol does not react appreciably at room temperature.

Victor Meyer's Test :–

The test consists of the following steps :

(i) The given alcohol is first converted into its alkyl iodine by treating it with phosphorus and iodine (P/I_2).

(ii) The alkyl iodide is then treated with silver nitrite ($AgNO_2$) to convert it into its corresponding nitroalkane.

(iii) The nitroalkane is then treated with nitrous acid (HNO_2) i.e. $NaNO_2$ + HCl.

(vi) The resulting solution is finally made alkaline with aq. NaOH or KOH.

Observations

- If blue colouration appears, it indicates a secondary alcohol.
- If the solution remains colourless, it indicates a tertiary alcohol.

Absolute Alcohol:

Denature Alcohol :

It is the alcohol rendered undrinkable by the addition of small quantities of benzene, toxic methanol.

Proof used to describe the alcohol content of wines and spirits

Proof is defined as twice the percentage by volume of alcohol in an ethanol water mixture. Thus a wine with alcohol 12% by volume is 24 proof.

Power Alcohol :–

The demand for gasoline (or petrol) for our daily life is increasing everyday. As the reserve of petroleum is limited, alternative sources of energy are needed. It has been found that absolute alcohol mixed with gasoline and benzene can be used as a motor fuel. Alcohol thus used for generating power is called power alcohol. It consists of a mixture of absolute alcohol and petrol roughly in the ratio 20 : 80. Since alcohol itself, does not mix with petrol, therefore a third solvent such as benzene or tetrahydronapthalene is added as a co-solvent.

Let's understand

Absolute alcohol is pure anhydrous ethanol. Since ethanol form a lower boiling azeotropic mixture containing 95.6 : 4.4 (v%) alcohol: water it cannot be dried by distillation. The water can be removed chemically by treatment with Mg turnings, which react with water forming $Mg(OH)_2$ and releasing H_2 gas. Alternatively, it can be distilled after the addition of some benzene. The benzene forms a ternary azeotrope with the alcohol water mixture which boils at 65ºC, lower than the b.p. of the binary azeotrope. Thus it distills first, removing the water and leaving behind water free alcohol, which is further distilled. Absolute alcohol purified in this manner should not be drunk because it contains small amounts of toxic carcinogenic benzene.

% Ionic Character Calculation -

$$\% \text{ ionic character} = \left(\frac{\text{electronegativity of C} - \text{elecctronegativity of metal}}{\text{electronegativity of C}}\right) \times 100$$

Example : $CH_3MgBr = \frac{2.5 - 1.2}{2.5} \times 100 = 50\%$ Ionic, 50% Covalent

Addition Reactions of Grignard Reagent

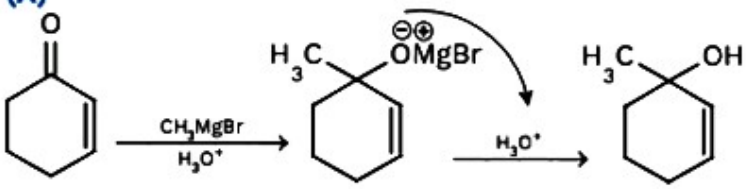
Important Concept - Weak nucleophile attack at π bond

Points to remember

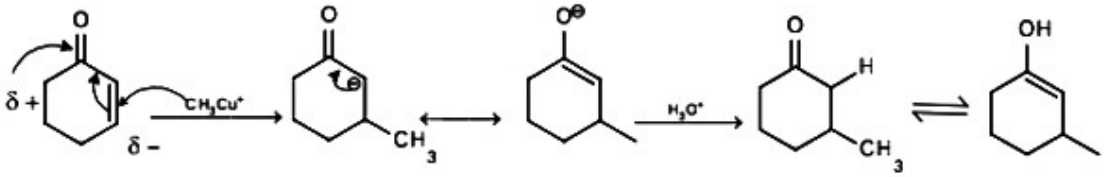
High potential/high charge density $E^{\oplus}$ prefer high potential $Nu^{\ominus}$ it means high charge density $E^{\oplus}$ reacts rapidly with high charge $Nu^{\ominus}$

Application

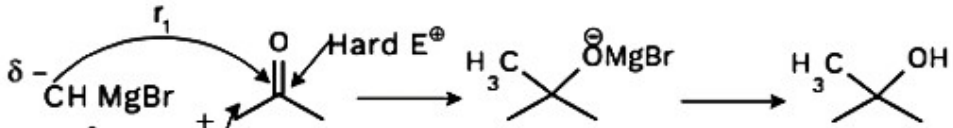
(A)



(B)



(C)



$r_1 > r_2$

Weak Nucleophile

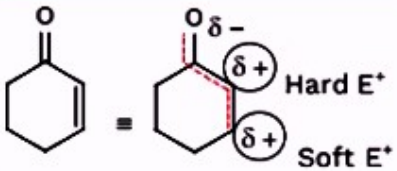
Grignard Reagents (R–Mg–X):

Preparation :

$$R\text{–}X + Mg \xrightarrow[\text{ether}]{\text{Reflux}} R\text{–}Mg\text{–}X$$

- $CH_3\text{–}Br \xrightarrow[\text{ether}]{\text{Reflux}} CH_3MgBr$
- $Ph\text{–}Br \xrightarrow[\text{ether}]{\text{Reflux}} PhMgBr$

JEE Concept



Mechanism :

(A) Ionic :

$$Mg \longrightarrow Mg^{2+} + 2e^-$$

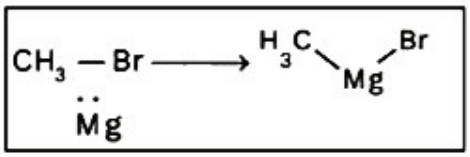
$$\overset{\ominus}{C}H_3\,\overset{\ominus}{Br} \xrightarrow{Mg^{2+}} \overset{\ominus}{C}H_3 - \overset{2+}{M}g\,\overset{\ominus}{Br}$$

Ionic

$$\overset{+}{C}H_3 \xrightarrow{+e^-} \overset{\bullet}{C}H_3 \xrightarrow{+e^-} \overset{\ominus}{C}H_3$$

$6e^-$ $7e^-$ $8e^-$

(B) Covalent :



(C) SET → Single electron transfer:

$$R - Br + \dot{M}g \longrightarrow R - \dot{B}r + M\dot{g}$$

$$R - B\dot{r} \longrightarrow \dot{R} + Br^-$$

$$Br^- + M\dot{g} \longrightarrow Br - M\dot{g}$$

$$Br - M\dot{g} + \dot{R} \longrightarrow R - Mg - Br$$

Introduction

The alkyl magnesium halides are extremely valuable inorganic synthesis.

Grignard reagents are prepared in the laboratory by the action of alkyl halides on magnesium metal in the presence of dry ether. Dry ether is pure anhydrous diethyl ether.

Q3 **Why R–Mg–X is highly unstable?**

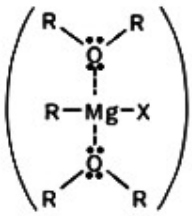
A3 It has less electronegativity but more electron donor. That's why it is highly unstable/reactive.

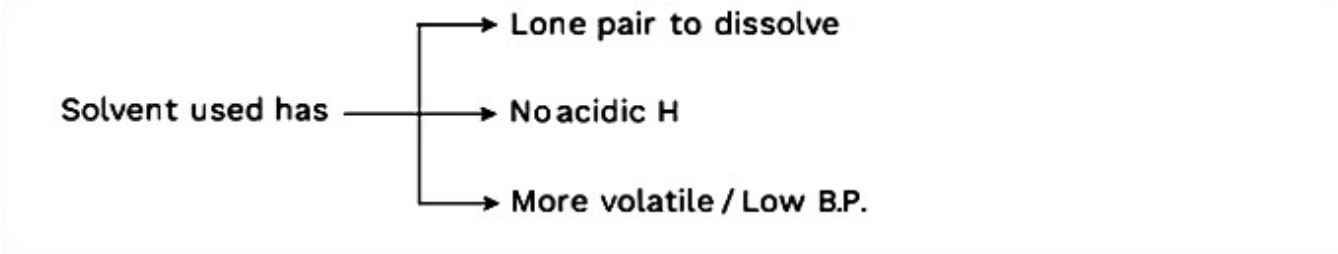
Q4 **How can we store highly unstable R–Mg–X?**

A4 (A) A Grignard reagent forms sigma complex with its ether solvent by making co-ordinate bond with two ether oxygen's (having lone pair). So that it can (re-main) dissolve in solution.

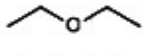
(B) Solvent must have low boiling point and is more volatile.

(C) Do not use polar protic solvent





Example:

(a)  diethylether (DEE) B.P. 35°C

(b)  Tetrahydro furan (THF) 60°C

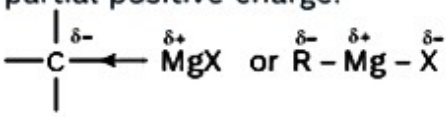
(c)  Pyridine 78°C

(d) R_3N

Properties of Grignard Reagent

Chemical properties of Grignard reagent:

- The carbon magnesium bond in Grignard reagents is covalent but highly polar. The carbon atom is more electronegative than magnesium. The electrons of C–Mg bond are drawn towards the carbon atom. As a result, the carbon atom has a partial –ve charge and the magnesium atom has a partial positive charge.



NOTE : R–MgX is highly reactive, Grignard reagents will also react with oxygen, carbon dioxide and water thus they are highly unstable when exposed to atmosphere.

Reactions of Grignard Reagent

(1) Acid-Base
(2) Nucleophilic addition
(3) Substitution (Addition – Elimination)
(4) SN_{AE}

Points to remember

Physical:

- Grignard reagents are non-volatile, colourless solids.
- They are seldom isolated in the free state on account of their explosive nature for synthetic purposes, they are always prepared and used in ether solution.

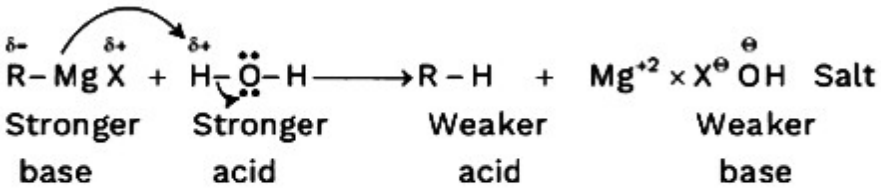
Points to remember

For a variety of reasons, anhydrous diethyl ether is the solvent of choice for carrying out a Grignard synthesis. Vapours from highly volatile solvent help to prevent oxygen from reaching the reaction solution

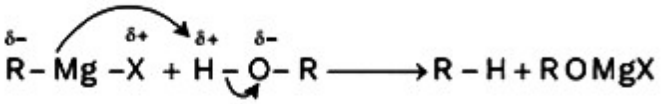
Acid-Base reaction of Grignard reagent:

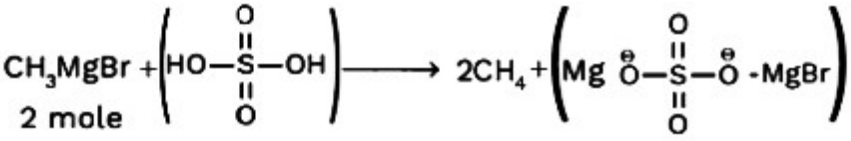
- Grignard reagents are very strong bases. They react with any compound that has a hydrogen atom attached to an electronegative atom such as oxygen, nitrogen and sulphur.
- Reactions of Grignard reagents with water and alcohols are nothing more than acid-base reactions; they lead to the formation of the weak-conjugate acid and weak conjugate base.

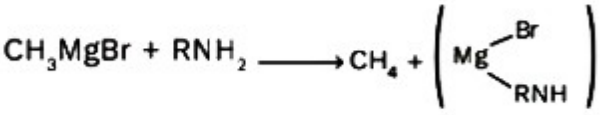
Example

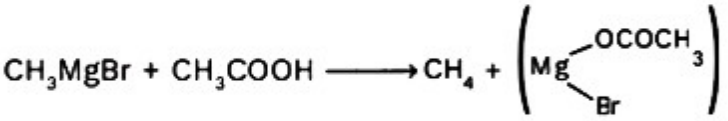


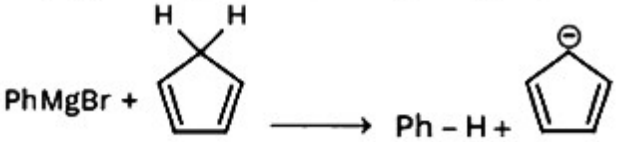
1. $CH_3MgBr + H-O-H \longrightarrow CH_3-H + HOMgBr$

2. 

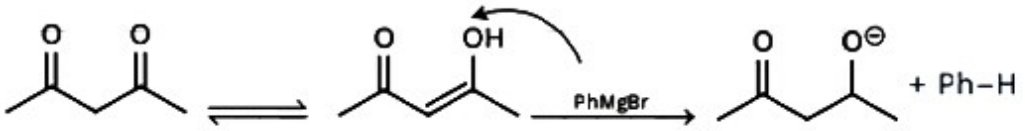
3. 

4. 

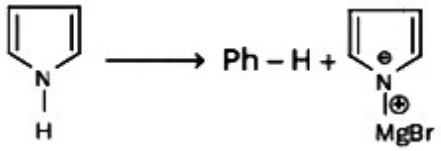
5. 

6. $CH_3MgBr + NH_3 \longrightarrow CH_4 + Mg(NH_2)Br$
7. $CH_3MgBr + HCl \longrightarrow CH_4 + Mg(Cl)Br$
8. $CH_3MgBr + H_2S \longrightarrow CH_4 + Mg(SH)Br$
9. 

10. $PhMgBr + R-C\equiv C-H \longrightarrow Ph-H + (Ph-C\equiv C)MgBr$
11. $PhMgBr + R-C\equiv C-CH_3 \longrightarrow$ Not possible.

12. PhMgBr + 

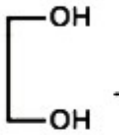
+ Ph–H

13. PhMgBr + 

14. Mole concept - ZereWitinoff Method

(a) $CH_3MgBr + CH_3CH_2OH \longrightarrow CH_4 + EtOMgBr$

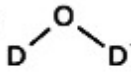
1 mole + 1 mole $\longrightarrow$ 1 mole = 16 gm = 22.4L

(b) $CH_3MgBr +$ 

$\longrightarrow CH_4$

2mole 1.2

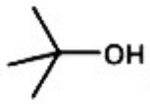
0.6

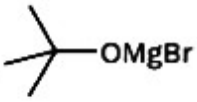
15. PhMgBr + 

$\longrightarrow Mg(\overset{\ominus}{O}D)Br^{\ominus}$ + Ph–D

16. $PhMgBr + CH_3OH \longrightarrow Ph–H + MgBr(O^-CH_3)$

17. $PhMgBr + PhOH \longrightarrow Ph–H + PhOMgBr$

18. PhMgBr + 

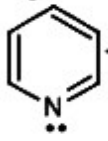
$\longrightarrow$ Ph – H + 

20. $PhMgBr + NH_3 \longrightarrow Ph–H + NH_2MgBr$

21. $PhMgBr + R–NH_2 \longrightarrow PhH + RNHMgBr$

22. $PhMgBr + R_2NH \longrightarrow PhH + R_2NMgBr$

23. $PhMgBr + R_3N \longrightarrow$ No Reaction (No active 'H' present)

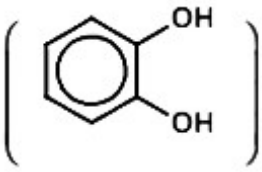
24. PhMgBr + 

$\longrightarrow$ No reaction

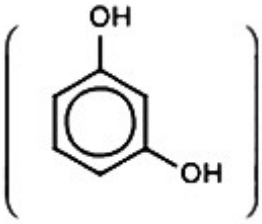
25. PhMgBr +  $\longrightarrow$ No reaction

26. PhMgBr +  $\longrightarrow$ No reaction

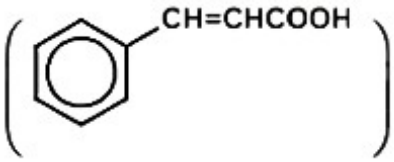
27. $CH_3MgBr + Catechol \longrightarrow 2CH_4$



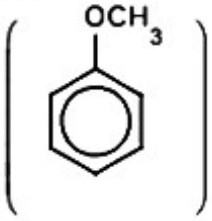
28. CH_3MgBr + Resorcinol $\longrightarrow$ $2CH_4$ + salt



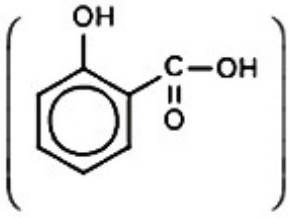
29. CH_3MgBr + Cinnamic acid $\longrightarrow$ CH_4 + salt



30. CH_3MgBr + Anisole $\longrightarrow$ No reaction

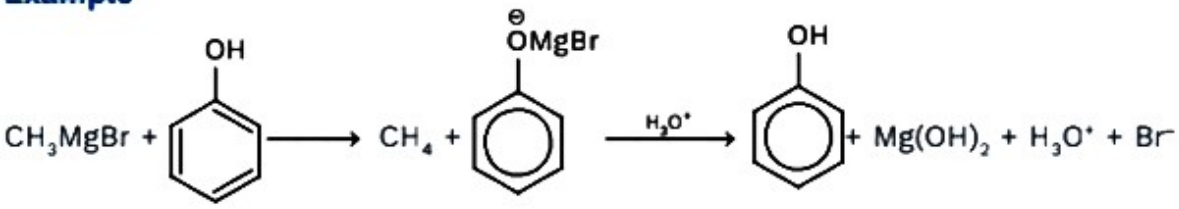


31. CH_3MgBr + Salicylic acid $\longrightarrow$ $2CH_4$



Do you know Why acid-base reaction is a wastage for R–MgX?
Reason - After hydrolysis, salt again form compound.

Example

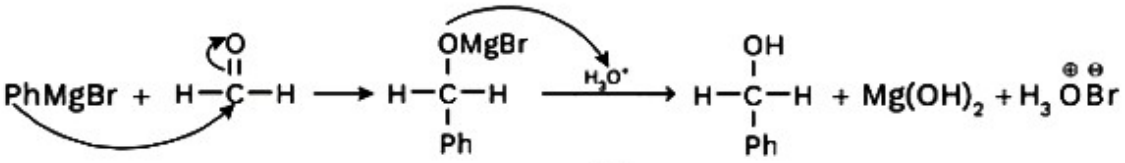


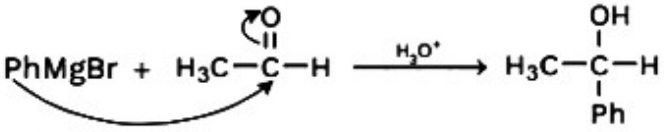
Nucleophilic Addition

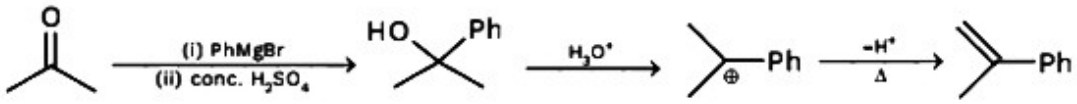
Carbonyl compounds are attacked by Grignard reagents to from addition products which upon acid hydrolysis yield alcohols, acids esters etc.

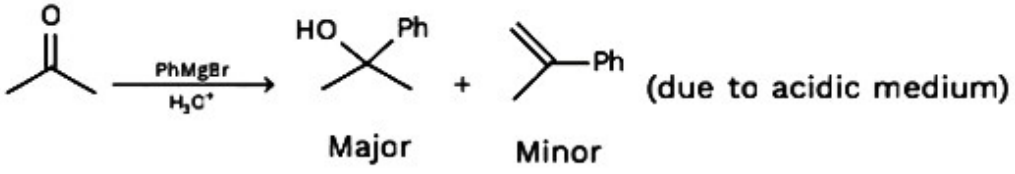
Order of reactivity of functional group

$-CHO > CO > -COCl > -CO_2R > -CH_2R$

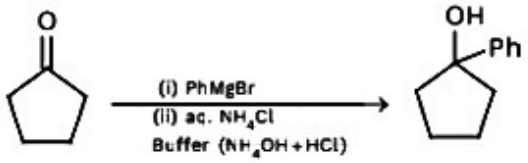
1. 

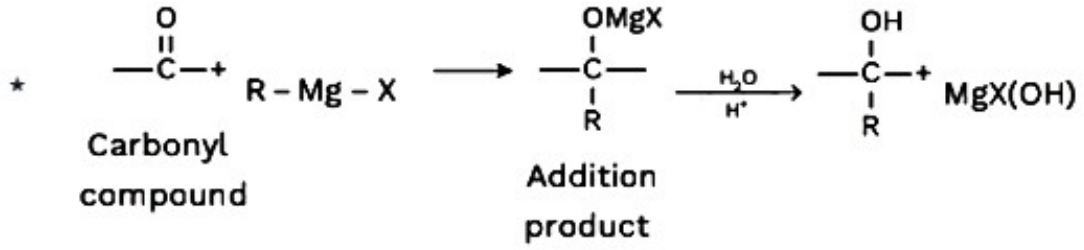
2. 

3. 

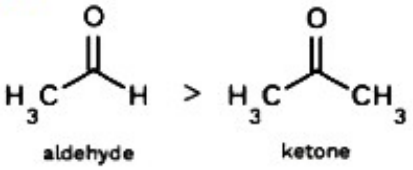
4. 

(due to acidic medium)

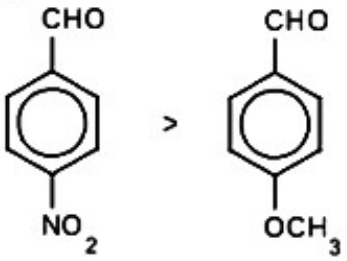
5. 



Rate of Order:



(Less steric + more + ve magnitude due to less + I)

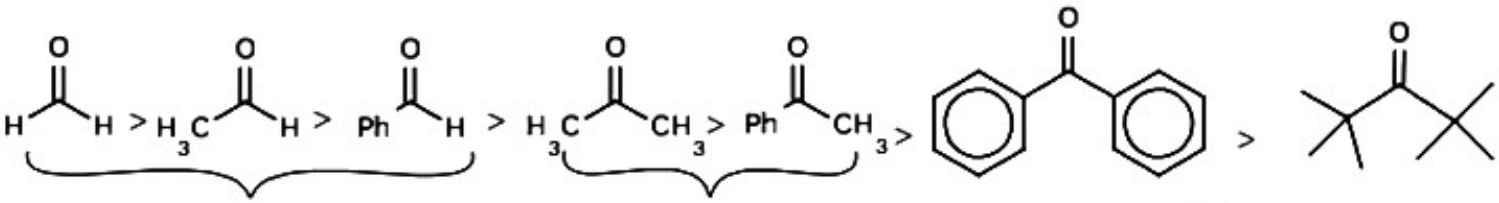


(–M) (+M)

Intra > Inter

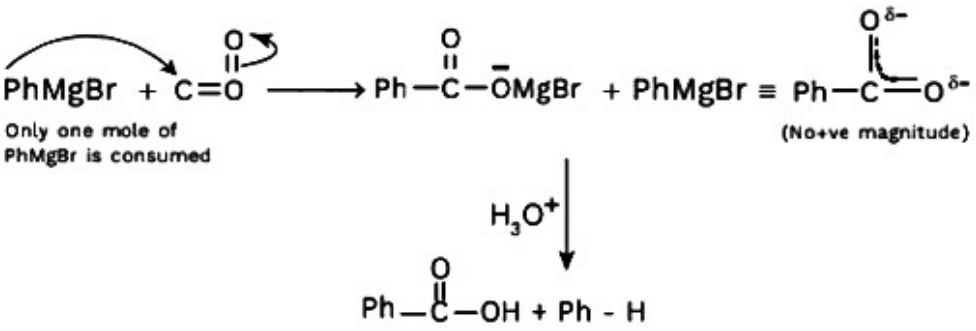
Let's understand

Grignard reagents react with a variety of substances yielding almost the entire range of organic substances

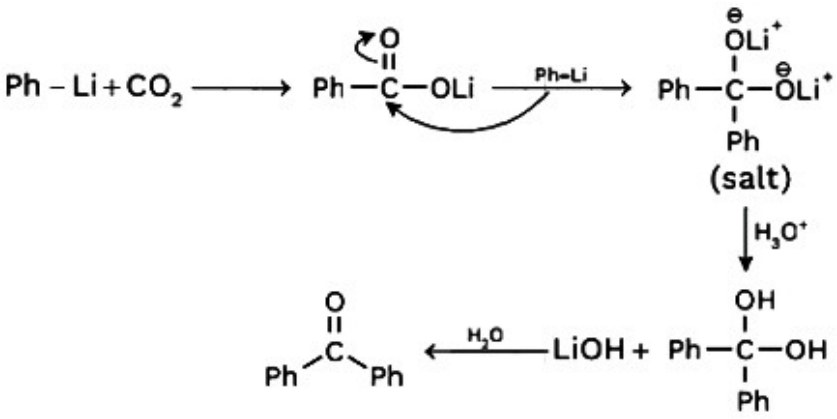


(Not give nucleophilic addition reaction due to steric crowd)

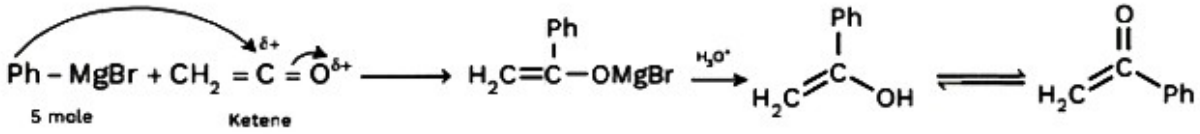
6.

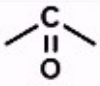


7.

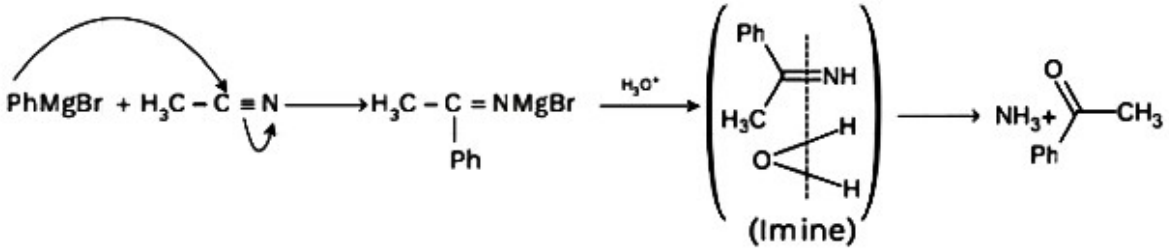


8.

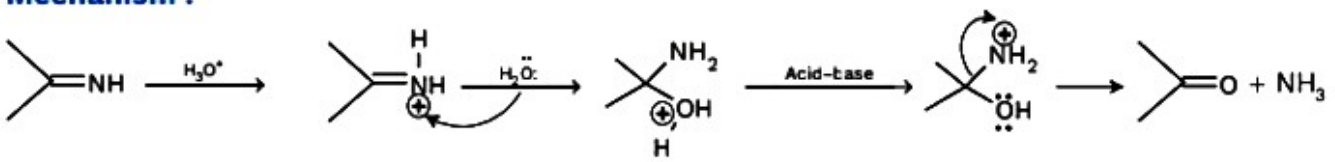


If once get hydrolysed,  bond do not further give nucleophilic addition reaction.

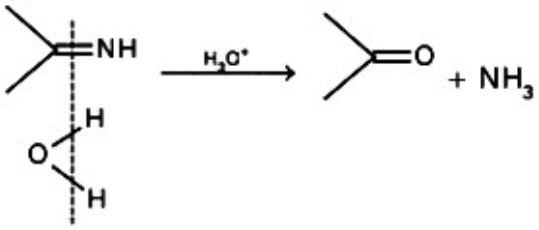
9.

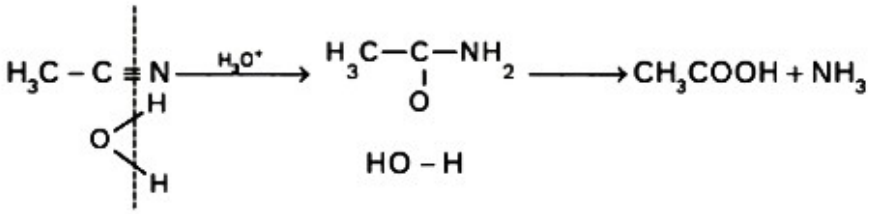


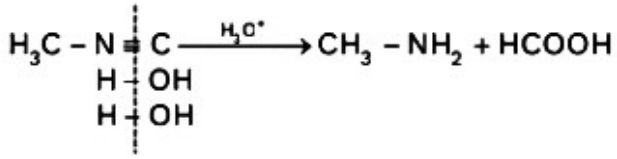
Mechanism :



Hint :

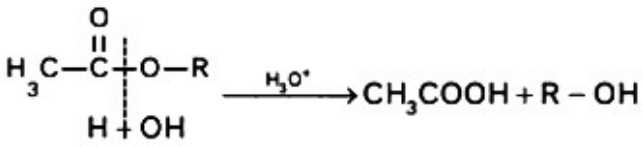
(A) 

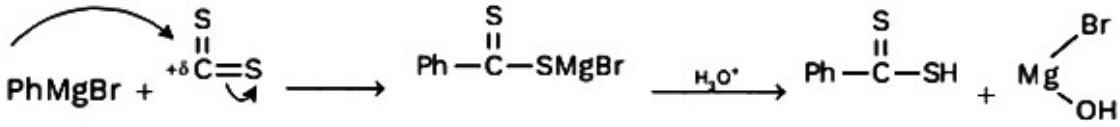
(B) 

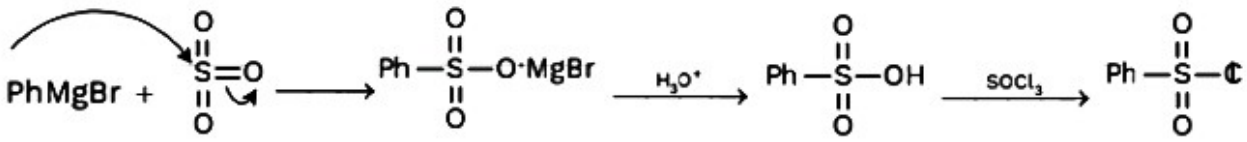
(C) 

Points to remember

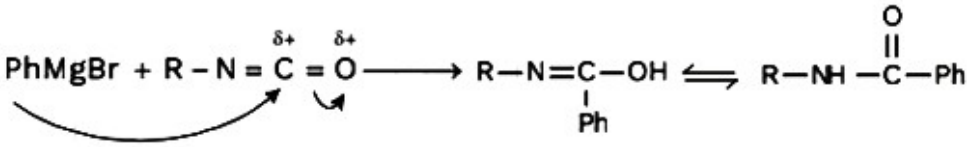
- Imide, cyanide, isocyanide and ester are unstable in acidic medium.

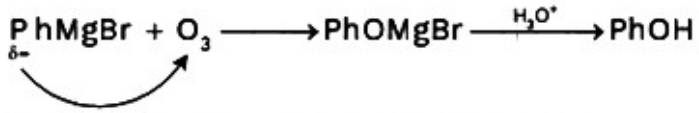
(D) 

10. 

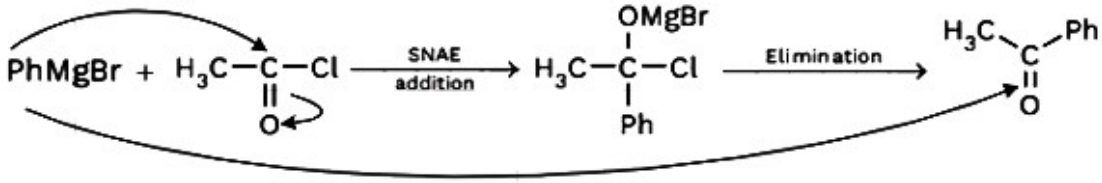
11. 

(Heinsberg reagent)

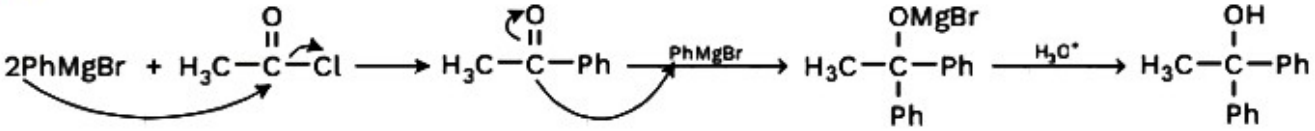
12. 

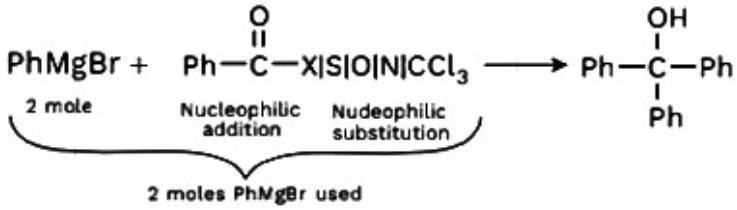
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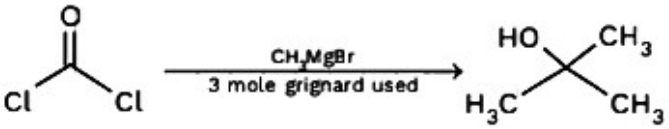
Substitution by Addition -Elimination Path

14. 

15.



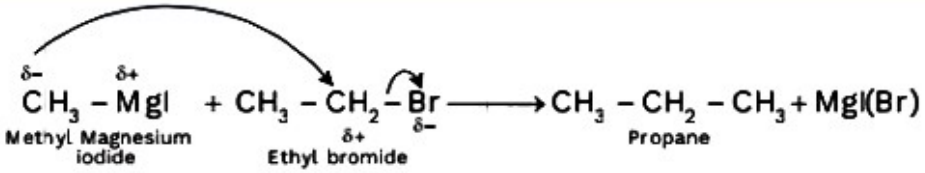
16. 

17. 

Reaction of Grignard reagent with alkyl halide:

Q5 $CH_3 - MgI$ (Methyl Magnesium iodide) $+ CH_3 - CH_2 - Br$ (Ethyl bromide)

A5



Q6 $CH_3MgI + CH_2 = CH-CH_2Br$

Allyl Bromide

A6 $CH_3MgI + CH_2 = CH-CH_2Br \longrightarrow CH_2=CH-CH_2-CH_3 + MgI(Br)$

Allyl Bromide

S_N2 is a one step process and involves no intermediate. In this reaction, the nucleophile attacks the substrate from the just opposite side to the leaving group.

Q7 $CH_3MgBr + CH_3-Br$

A7 $CH_3MgBr + CH_3-Br \longrightarrow CH_3-CH_3 + MgBr$

Q8 $PhMgBr + CH_3-\ddot{O}-Cl$

A8 $PhMgBr + CH_3-\ddot{O}-Cl \longrightarrow$ No reaction

Q9 $PhMgBr + H_2C(OCH_3)(Cl)$

A9 $PhMgBr + H_2C(OCH_3)(Cl) \longrightarrow Ph-CH_2-OCH_3 + Mg(Br)Cl$

Q10 $CH_3MgBr + I-Br$

A10 $CH_3MgBr + I-Br \longrightarrow CH_3-I + MgBr_2$

Q11 $CH_3MgBr + Br-Cl$

A11 $CH_3MgBr + Br-Cl \longrightarrow CH_3Br + Mg(Br)Cl$

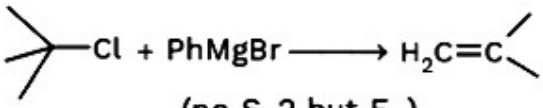
Q12 **$CH_3MgBr + CH_3–Cl$**

A12 $CH_3MgBr + CH_3–Cl \longrightarrow CH_3–CH_3 + Mg(Br)Cl$

Q13 **$CH_3MgBr + CH_3–O–CH_2–Cl$**

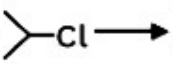
A13 $CH_3MgBr + CH_3–O–CH_2–Cl \longrightarrow CH_3–O–CH_2–CH_3$

Q14  **+ PhMgBr**

A14 

(no S_N2 but E_2)

Q15 **PhMgBr +** 

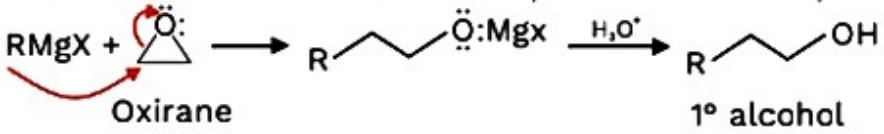
A15 PhMgBr +  +  (both E_2 , S_N2)

Q16 **PhMgBr + Br–Br**

A16 $PhMgBr + Br–Br \longrightarrow PhBr + MgBr_2$

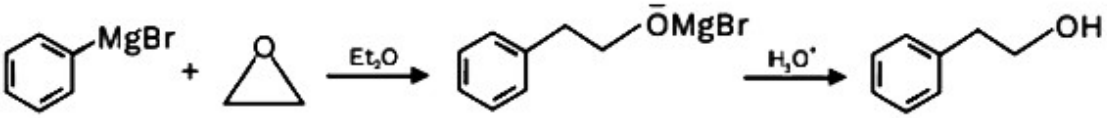
Reaction of Grignard with epoxide :

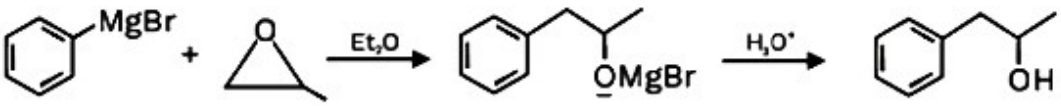
The nucleophilic alkyl group of the Grignard reagent attacks the partially positive carbon of the epoxide ring. Because it is rightly strained, the ring opens, and the reaction leads to the alkoixde salt of an alcohol. Subsequent acidification produces the alcohol.

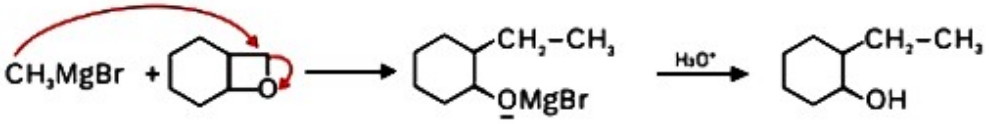


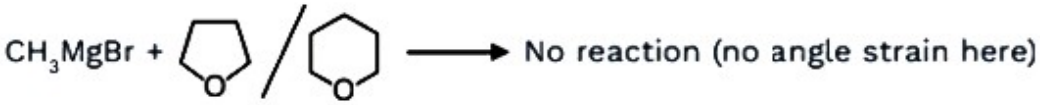
Oxirane

1° alcohol





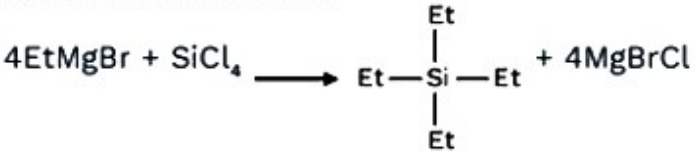




No reaction (no angle strain here)

Transmetallation : (Transfer of Metal)

Tramsmetallation is a type of organometallic reaction that involves transfer of ligands from one metal to another.



$3C_2H_5MgBr + AlCl_3 \longrightarrow (C_2H_5)3Al + 3MgBrCl$

$4C_2H_5MgBr + TiCl_4 \longrightarrow Ti(C_2H_5)_4 + 4\ MgBrCl$

$2C_2H_5MgBr + ZnCl_2 \longrightarrow (C_2H_5)_2Zn + 2\ MgBrCl$

$2C_2H_5MgBr + CuCl_2 \longrightarrow 2[(C_2H_5)-Cu]Cl^- \longrightarrow (C_2H_5)_2Cu$

$4C_2H_5MgBr + PbCl_4 \longrightarrow (C_2H_5)_4Pb + 2MgCl_2 + 2MgBr_2$ (T.E.L)/Anti knocking agent

Tetraethyl lead

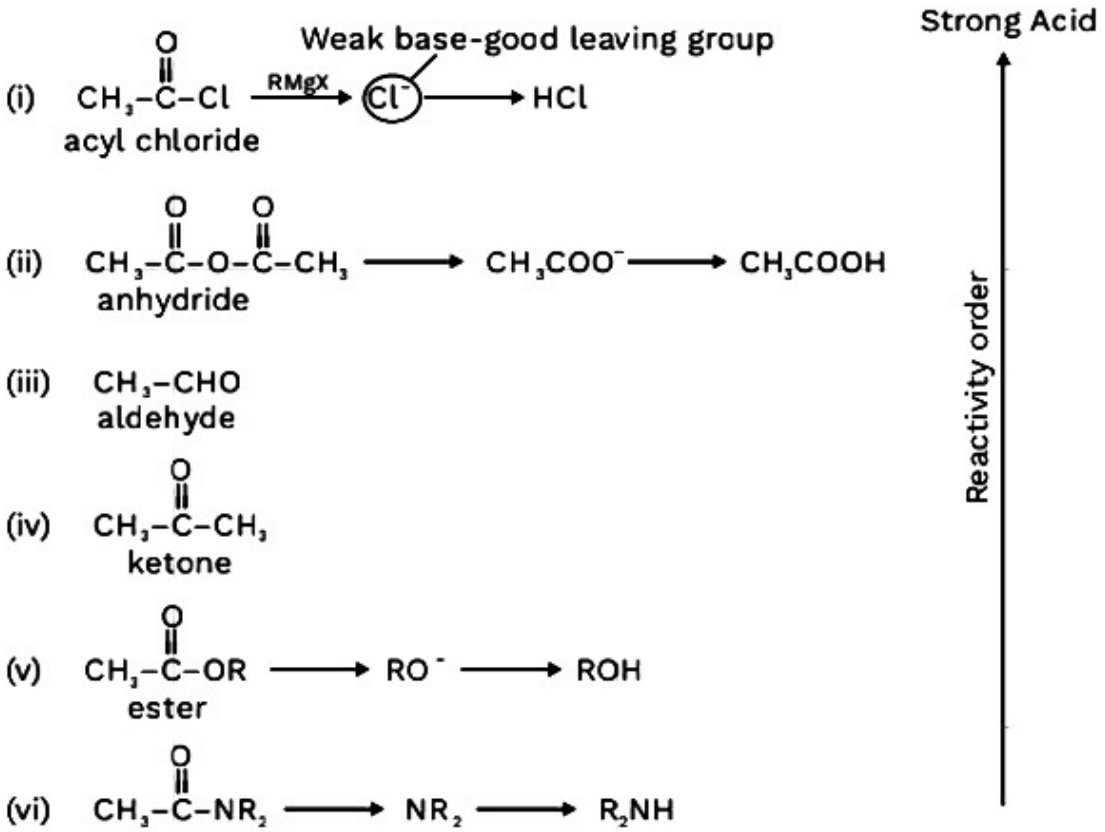
$2CH_3CH_2MgCl + CdCl_2 \longrightarrow (CH_3CH_2)_2Cd + 2MgCl_2$

Ethyl magnesium Chloride

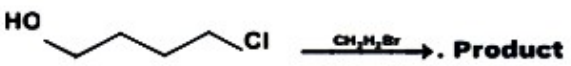
diethylcadmium

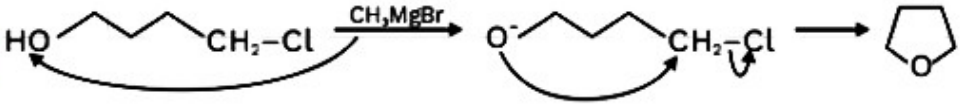
Reactivity order of R–Mg–X

Acid–base	>	Nucleophilic reaction	>	SN^2
(non-directional fast reaction)		(π^* attack low energy)		(σ^* high energy)

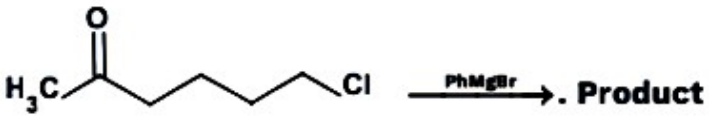


Solved Examples :

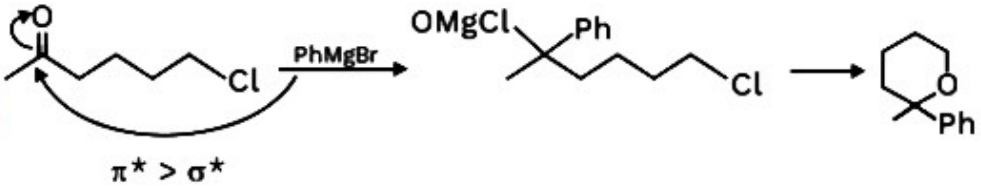
Q17 

A17 

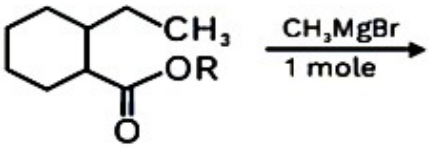
Q18



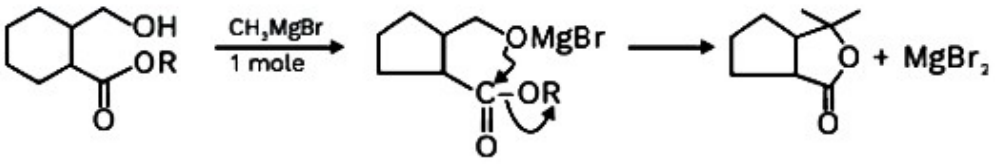
A18



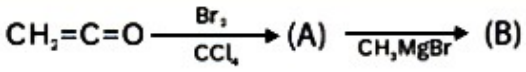
Q19



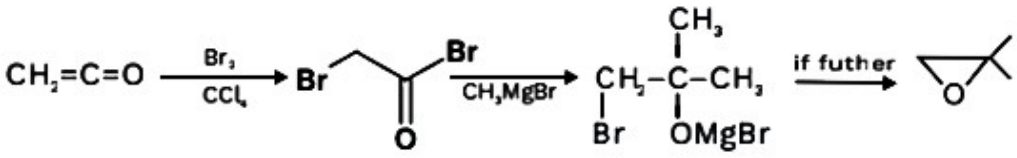
A19



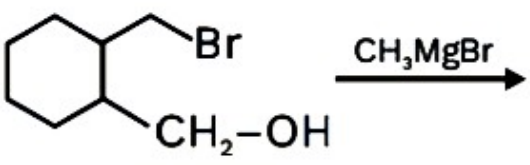
Q20



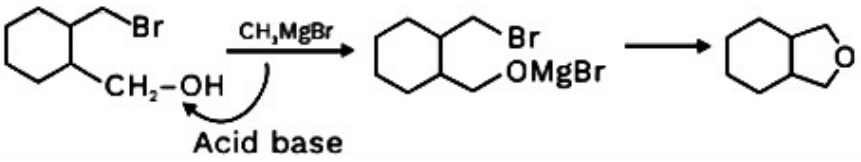
A20



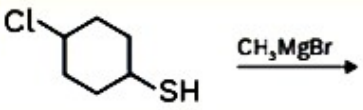
Q21



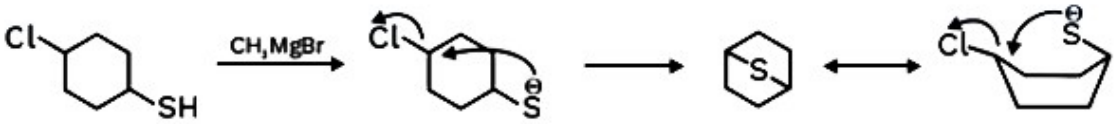
A21



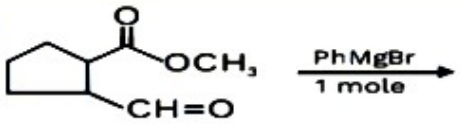
Q22



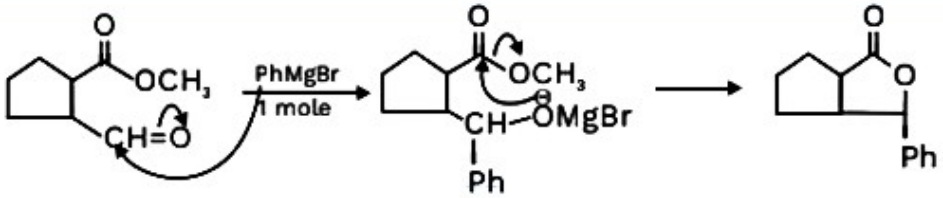
A22



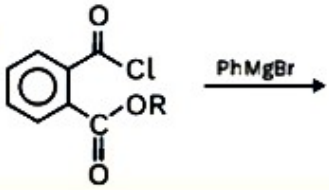
Q23



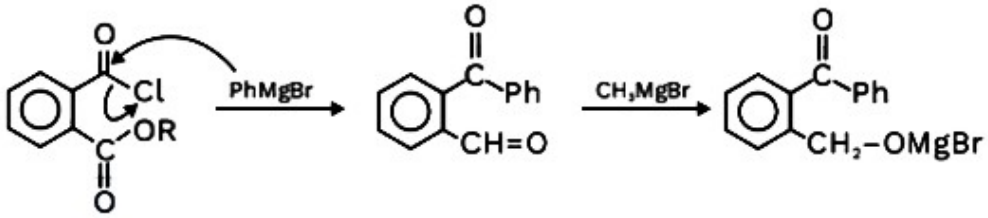
A23



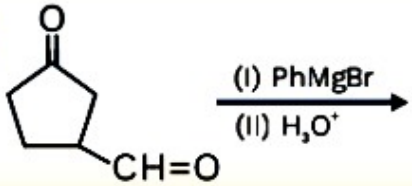
Q24



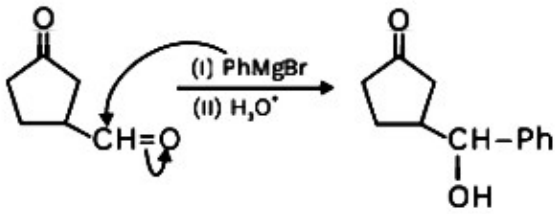
A24



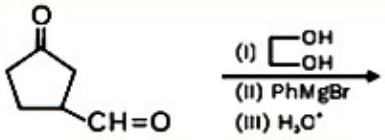
Q25



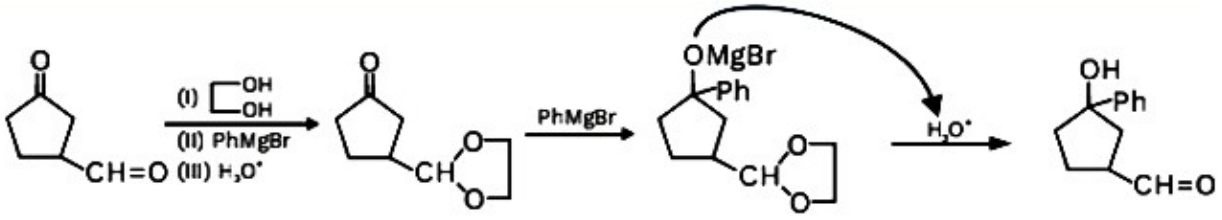
A25



Q26



A26

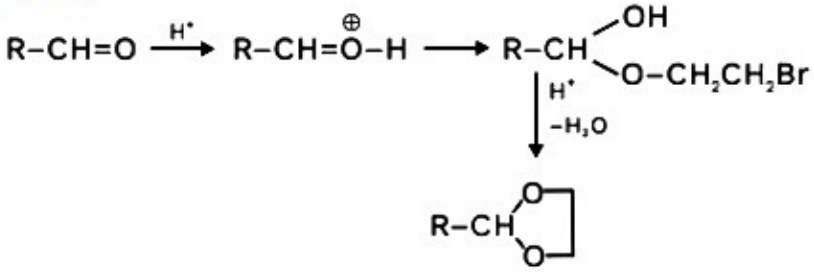


Acetal

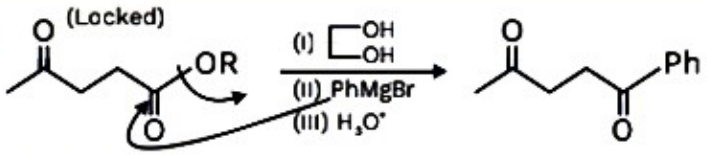
(React fast in presence of glycol)

Glycol is used to lock more reactive site :

Mechanism :

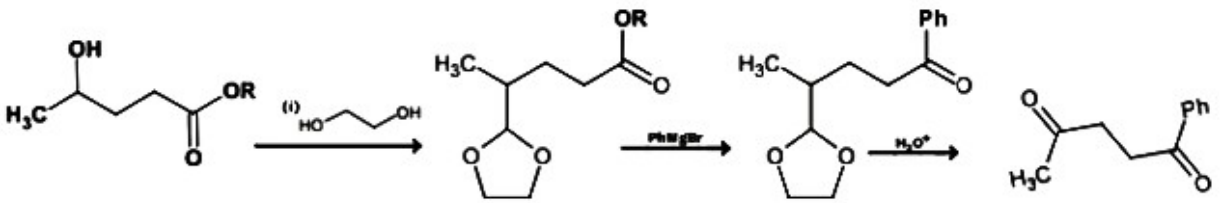


Q27



Glycol is used as protecting group for more reactive site by acetal formation → RMgX does not react with acetal

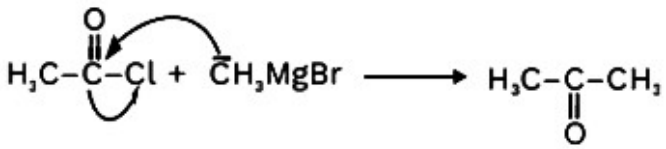
A27



Q28



A28



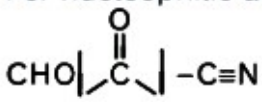
Tips and Trick

Number of moles consumed by R–Mg–X

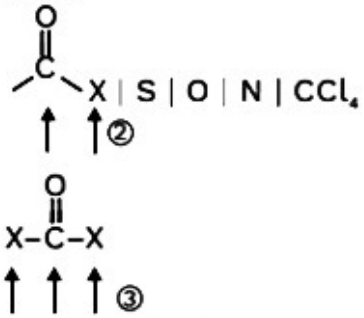
(A) Acid base → per mole RMgX
H–X | S | O | N | CCl_4
H–C → sp/aromatic/stable enol.

(B) per SN^2 → 1 mole.

(C) Per nucleophilic addition → 1 mole



(D) SNAE ®

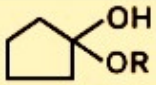


(E) Acetal, ether, Alkene, Internal alkyne, 3° amine → zero

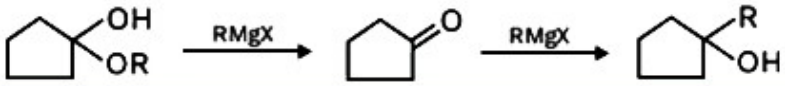
(F) Hemiacetal and anhydride – 2

Solved Examples :

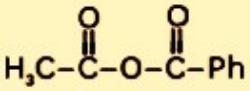
Q29



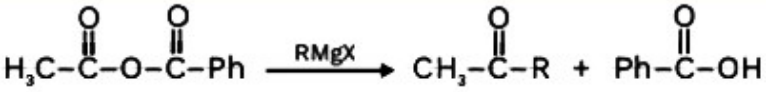
A29



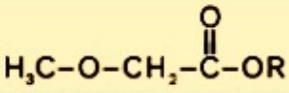
Q30



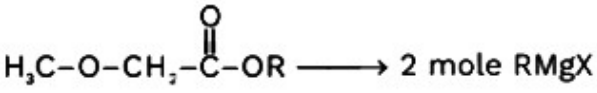
A30



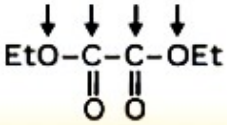
Q31



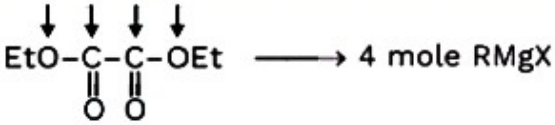
A31



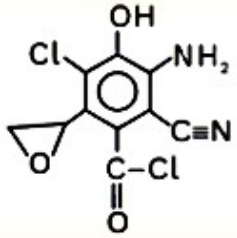
Q32



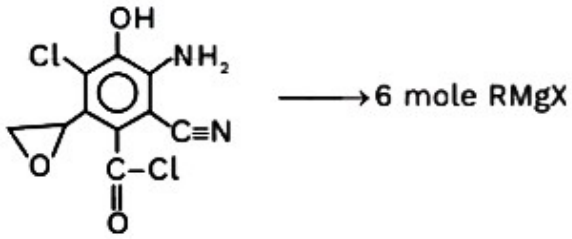
A32



Q33

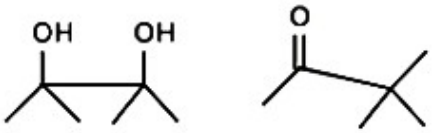


A33



Introduction of Pinacol and Pinacolone

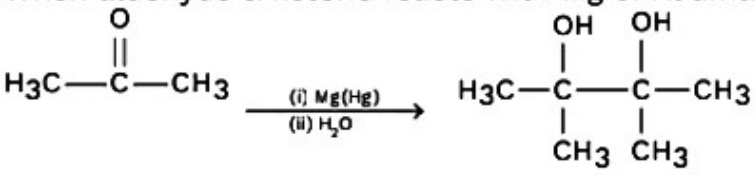
This reaction involves the reductive homo-coupling of a carbonyl compound to produce a symmetrically substituted 1,2-diol. The first step is single electron transfer of the carbonyl bond, which generates radical ion intermediates that couple via carbon-carbon bond formation to give a 1,2-diol. Pinacol and other highly substituted 1,2-diols tend to undergo dehydration with rearrangement under acid-catalysis :



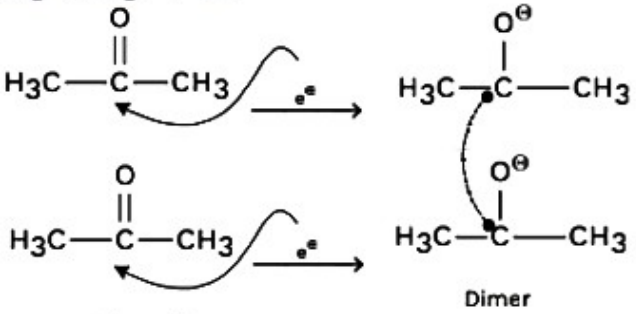
Pinacol Pinacolone

Preparation of pinacol

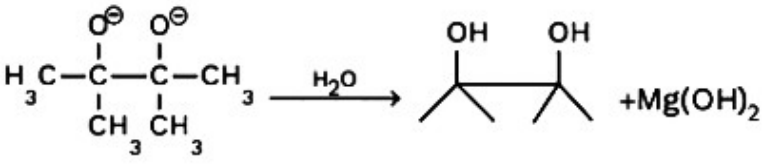
When aldehyde & ketone reacts with Mg or Al amalgum formation of pinacol takes place



$Mg \rightarrow Mg^{+2} + 2e^-$



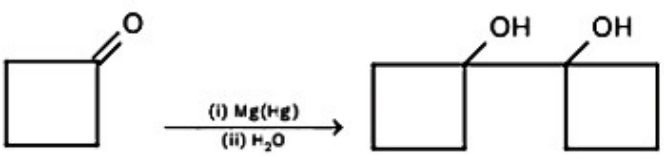
Dimer

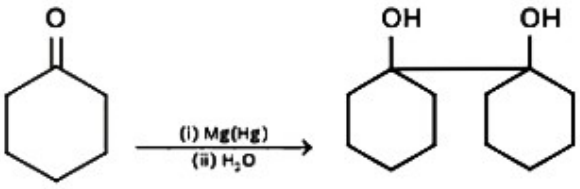


Characteristic of reaction

Free Radical is the intermediate, product is obtained in basic medium due to formation of base $Mg(OH)_2$.

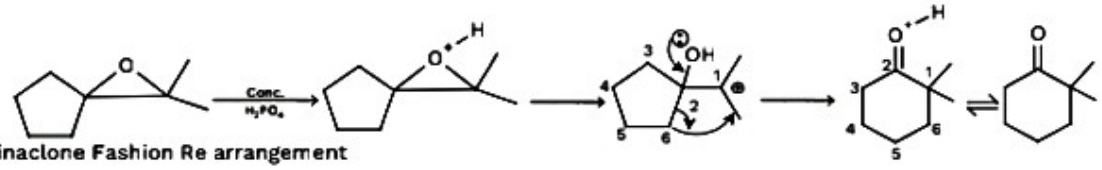
Example:





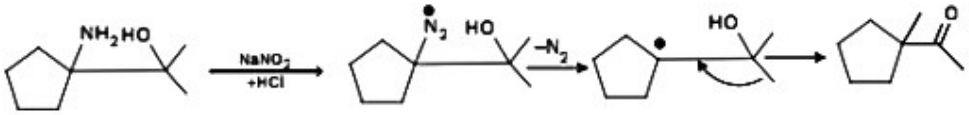
Pinaclone Fashion Rearrangement

(i)



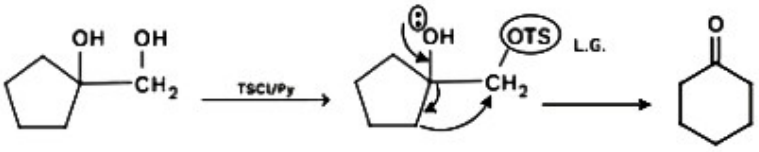
Pinaclone Fashion Re arrangement

(ii)

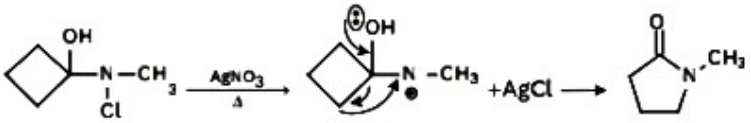


Pinacolone Didaotisation

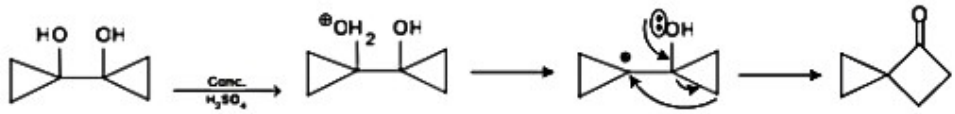
(iii)



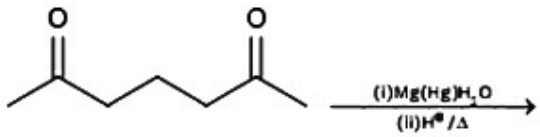
(iv)



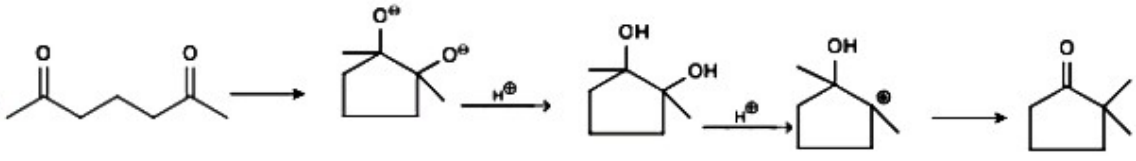
(v)



(vi)



Sol.



Characteristic of reaction

(i) Formation of carbocation is rds.

(ii) It is a four-step reaction.

(iii) Overall reaction is exothermic because of formation of C=O

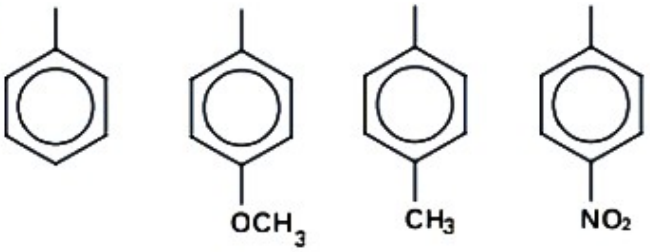
(iv) Migration in intramolecular.

(v) Migration aptitude

$-H > -Ph > 3°C > 2°C > 1°C > -CH_3$

Example :

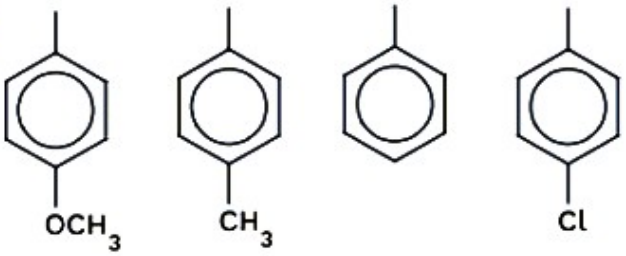
Q57



(A) (B) (C) (D)

A57 B > C > A > D

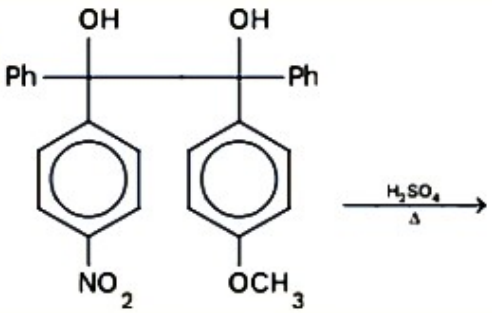
Q58



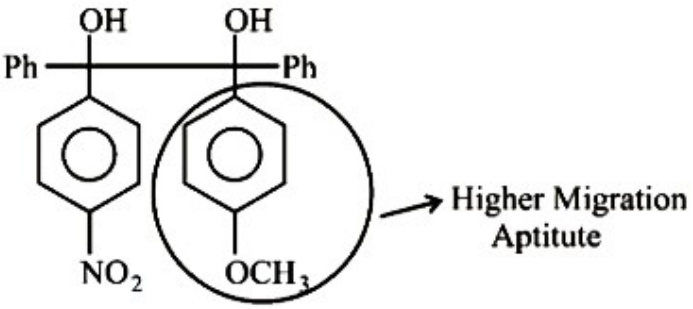
(A) (B) (C) (D)

Q58 A > B > C > D

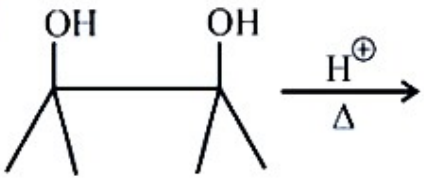
Q59



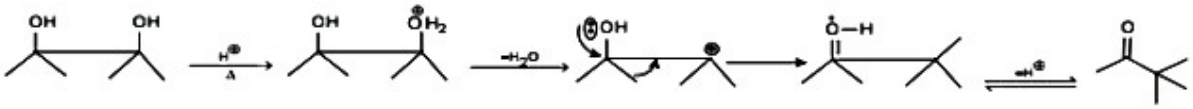
Q59



Q60



A60



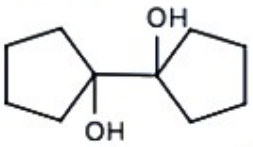
(Ketone)

Q61 **2-methyl-1,2,-propanediol is an unsymmetrical vicinal diol, how may product is obtained when it is dehydrated in an acidic solution.**

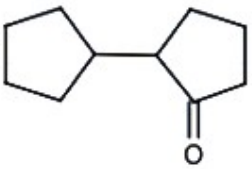
(A) 0 **(B) 1** **(C) 2** **(D) 3**

A61 (B)

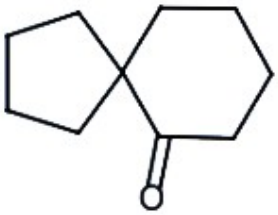
Q62 **What product is obtained when the following vicinal diol is heated in an acidic solution ?**



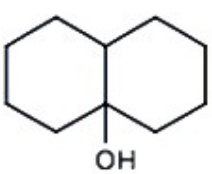
(A)



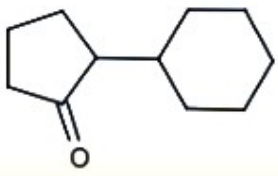
(B)



(C)

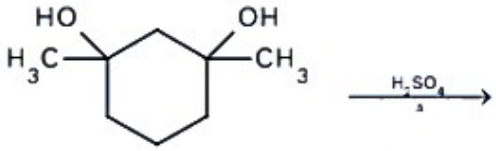


(D)



A62 (B)

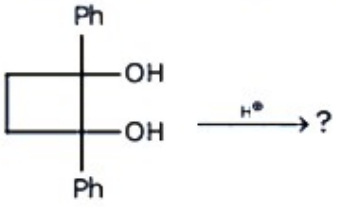
Q63 **Number of possible product/s of given vicinal diol after pinacol rearrangement**



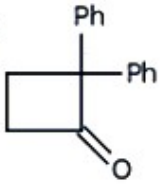
(A) 1 **(B) 2** **(C) 3** **(D) 4**

A63 (B)

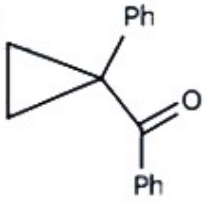
Q64 **Identify the product for the following reaction.**



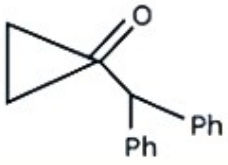
(A)



(B)



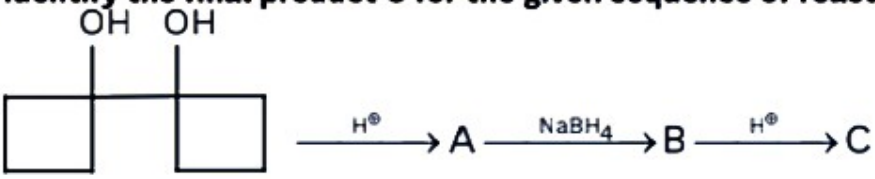
(C)



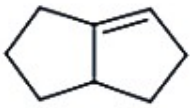
(D) No reaction

A64 (B)

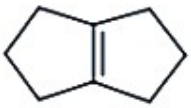
Q65 **Identify the final product C for the given sequence of reaction.**



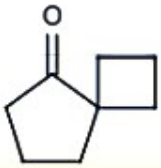
(A)



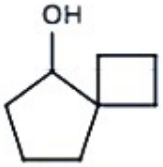
(B)



(C)

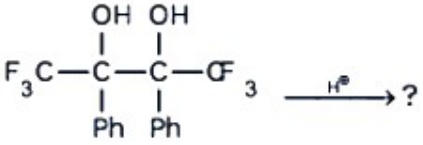


(D)

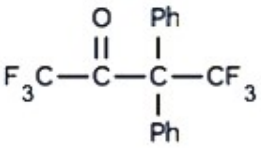


Q65 (B)

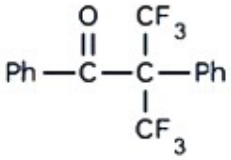
Q66 **For the following reaction suggest the suitable product.**



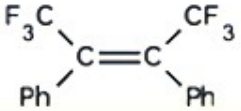
(A)



(B)



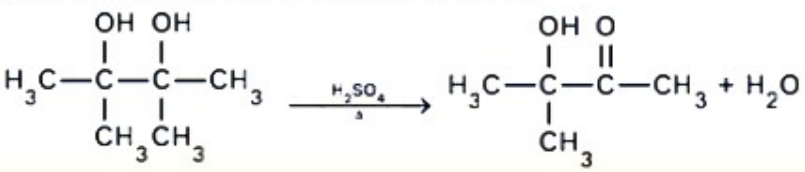
(C)



(D) No pinacol rearrangement

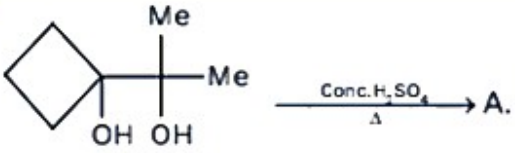
A66 (D)

Q67 **A vicinal diol has OH groups on adjacent carbons. The dehydration of a vicinal diol is accompanied by a rearrangement called to pinacol rearrangement. Propose a mechanism for this reaction.**



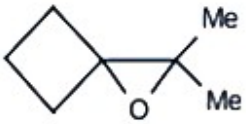
A67 (C)

Q68

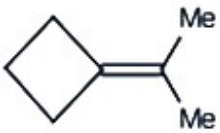


Product A is :

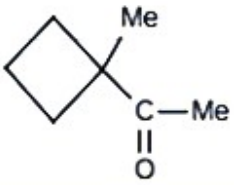
(A)



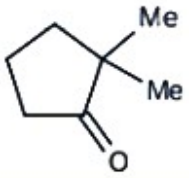
(B)



(C)

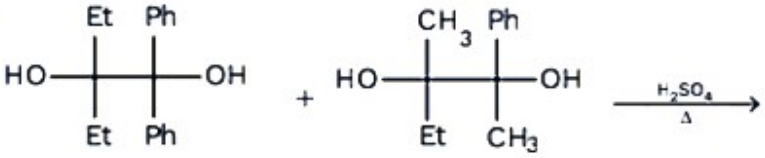


(D)



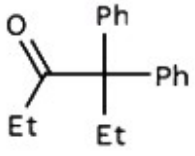
A68 (D)

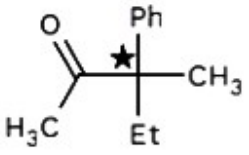
Q69 **How many products are obtained in the given reaction :**



(A) 1 **(B) 2** **(C) 3** **(D) 4**

A69 (D)





General Methods of preparation of Alcohol

1. Hydrolysis of alkyl halide :

Alkylhalides undergo hydrolysis when heated with aq. alkalies or moist AgOH the hydrolysis occurs by a nucleophilic substitution reaction

$$R - X + KOH(aq) \rightarrow R - OH + KX$$

$$CH_3 - CH_2 - Br + KOH(aq) \rightarrow CH_3 - CH_2 - OH + KBr$$

2. From Alkenes :

(a) Acid catalysed hydration of alkenes : Addition of water to an unsymmetrical alkene follows morkonikov's rule

$$H_3C - CH = CH_2 + H_2O \xrightarrow{H_2SO_4} H_3C - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_3$$

This method is suitable for the preparation of secondary and tertiary alcohols only.

(b) Oxymercuration – demercuration process (OMDM): In this method alkenes is treated with mercuric acetate, in presence of a mixture of water and THF as solvent (mercuration). The mercuration product is reduced with sodium borohydride to form alcohol (demercuration step). This reaction follows anti Markonikove's rule.

Example :

$$R - CH = CH_2 \xrightarrow[2.THF/H_2O]{1.\ Hg(OAc)_2} R-\underset{\displaystyle OH}{\underset{|}{CH}}-\underset{\displaystyle Hg(OAc)_2}{\underset{|}{CH_2}} \xrightarrow{NaBH_4} R-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3 + Hg + CH_3COO^-$$

$$CH_3 - CH_2 = CH_2 \xrightarrow[2.THF/H_2O]{1.\ Hg(OAc)_2} H_3C-\underset{\displaystyle OH}{\underset{|}{CH}}-\underset{\displaystyle Hg(OAc)_2}{\underset{|}{CH_2}} \xrightarrow{NaBH_4} H_3C-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3 + Hg + CH_3COO^-$$

Propan – 2 – ol

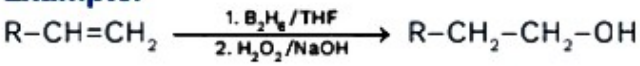
$$H_3C-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-CH=CH_2 \xrightarrow[(ii)\ NaBH_4]{(i)\ Hg(OAc)_2, H_2O} H_3C-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$$

3, 3 – dimethyl – 1 – butene

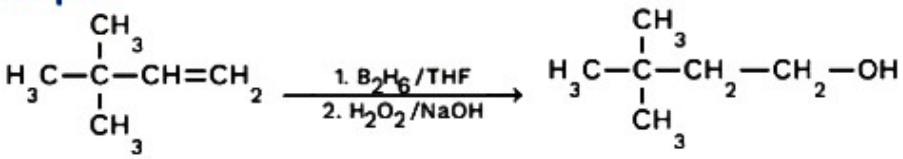
3, 3 – dimethyl – 2 – butanol

(c) Hydroboration-Oxidation : Alkenes react with borane in presence of ether to form alkyl boranes (hydroboration). Diborane is used in the form of solution in THF. Alkylboranes are then oxidised and hydrolysed by the addition of H_2O_2 in aqueous base to form alcohols.

Example:

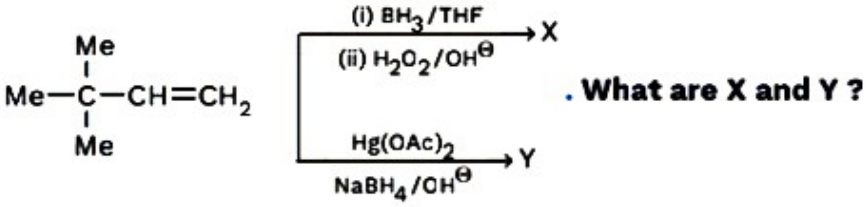


Example:



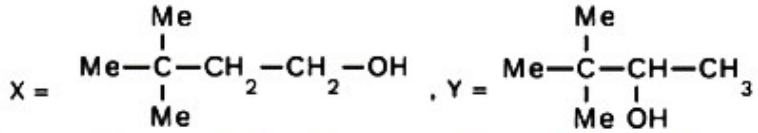
3, 3 – dimethyl – 1 – butene 3, 3 – dimethyl – 1 – butanol

Q70

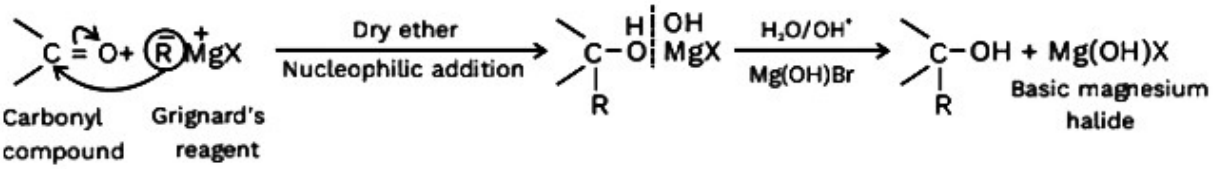


. What are X and Y ?

A70

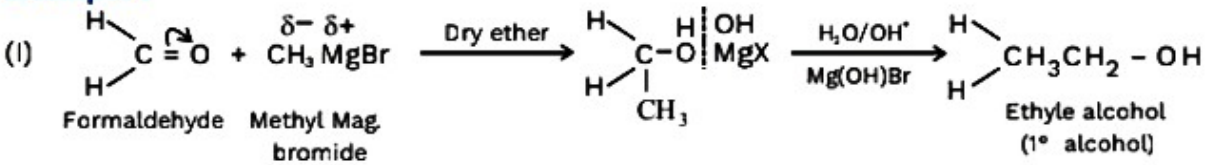


3. From primary carbonyl compound and Grignards reagent



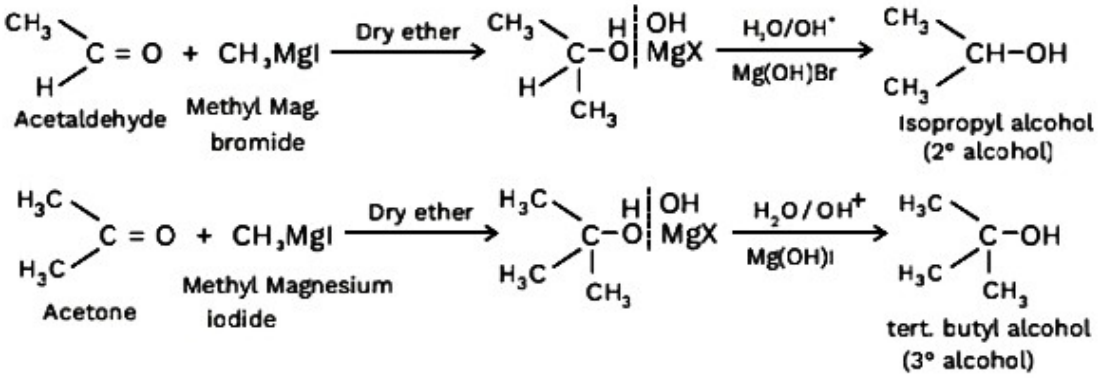
Carbonyl compound Grignard's reagent Basic magnesium halide

Example :



Formaldehyde Methyl Mag. bromide Ethyle alcohol (1° alcohol)

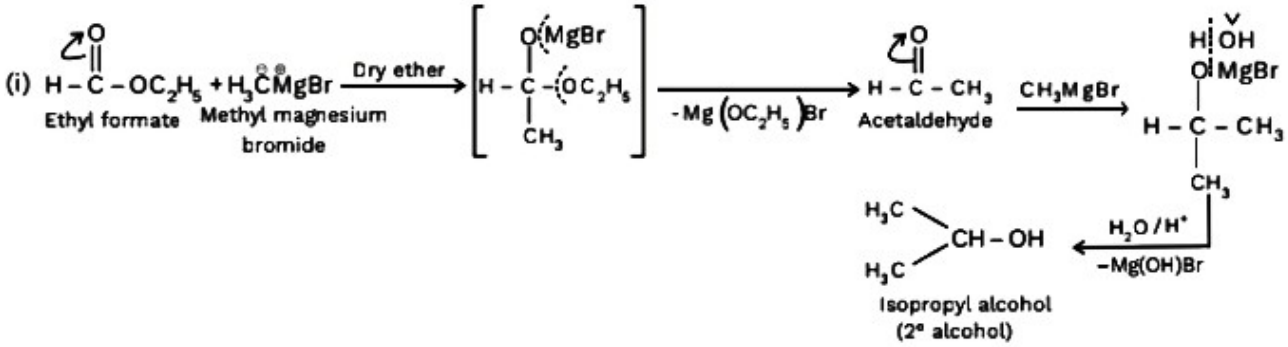
(ii)



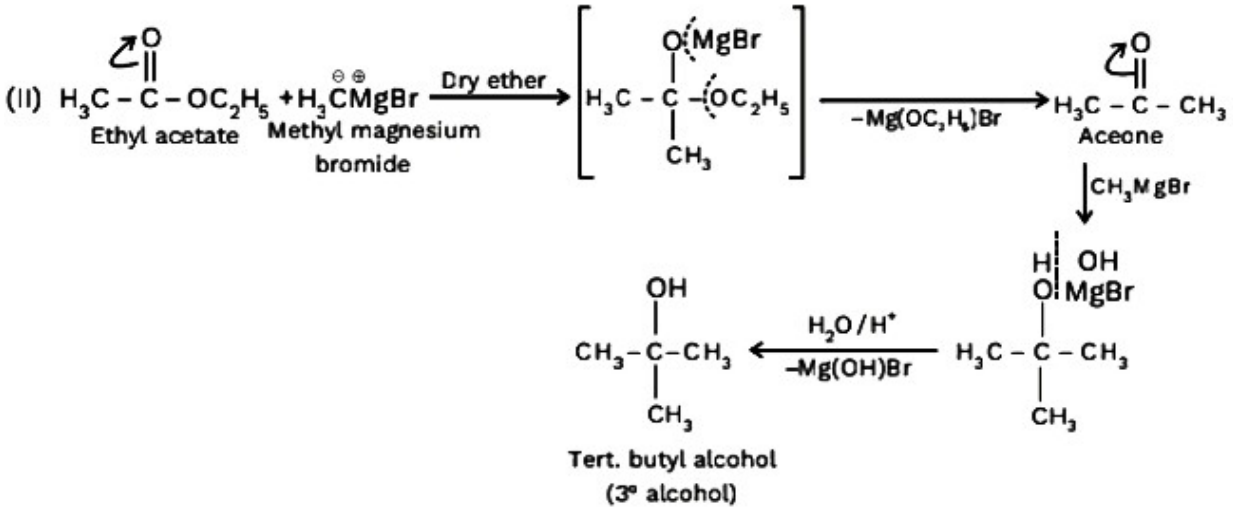
4. From Esters and Grignard reagents :

Ester of formic acid give 2° alcohol while ester of any other acid gives 3° alcohol.

(i)



(II)



5. By heating ethers with dilute sulphuric acid under pressure e.g. diethyl ether forms ethanol.

$$(C_2H_5)O + H_2O \xrightarrow{H_2SO_4} 2C_2H_5OH$$

6. By the reduction of aldehydes, ketones or esters by means of excess of sodium and ethanol or n-butanol as the reducing agent (Bouveault–Blanc reduction).

(i) Aldehydes : $R-CHO \xrightarrow{[H]} RCH_2OH$

(ii) Ketone : $R_2-CO \xrightarrow{[H]} R_2CHOH$

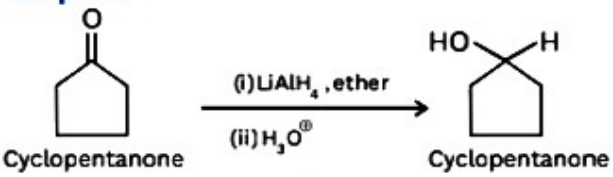
(iii) Ester :

$$R-\overset{\overset{O}{\|}}{C}-OR' \xrightarrow{[H]} RCH_2OH + R'OH$$

7. Reduction of Carbonyl Compounds :

Lithium aluminium hydride, $LiAlH_4$ is a versatile reducing agent which reduces aldehydes to primary alcohols and ketones to secondary alcohols. The reduction can also be done by catalytic hydrogenation. Such reduction is useful for the preparation of certain alcohols that are less available than the corresponding carbonyl compounds.

For Example :

(i) 

(ii) $CH_3CH=CHCHO \xrightarrow{H_2/Ni} CH_3CH_2CH_2OH$

Croton aldehyde — n–Butyl alcohol

(iii) $CH_3-CH=CH-CHO \xrightarrow[H_3O^+]{LiAlH_4,\ ether} CH_3-CH_2=CH-CH_2OH$

(iv) $C_6H_5-CH=CH-CHO \xrightarrow[H_3O^+]{NaBH_4,\ ether} C_6H_5-CH_2-CH_2-CH_2OH$

Cinnamaldehyde — 3-Phenyl propan-1-ol

8. Reduction of Acids to alcohols : LAH can also reduce an acid to an alcohol; the initial product is an alkoxide from which the alcohol is produced by hydrolysis :

$4\ RCOOH + 3\ LiAlH_4 \rightarrow 4\ RCH_2OH$ 1° al

9. Reduction of esters :

Like many organic compounds esters can be reduced in two ways :

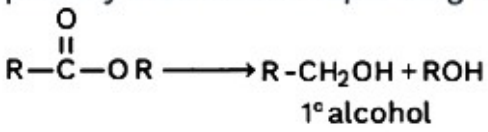
(a) by catalytic hydrogenation using molecular hydrogen or

(b) Chemical reduction. In either case, the ester is cleaved to yield (in addition to the alcohol from which it was derived) a

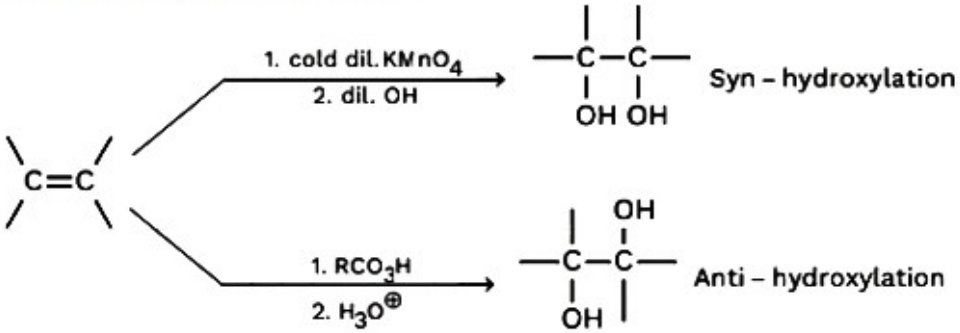
Points to remember

LAH cannot reduce C = C but when it is in conjugation then only it is reduced by LAH but sodium borohydride, $NaBH_4$, does not reduce carbon-carbon double bonds, not even those conjugated with carbonyl groups and in thus useful for the reduction of such unsaturated carbonyl compounds unsaturated alcohols. The reduction of the carbonyl group by LAH or $NaBH_4$ occurs in a stepwise manner, each step involving hydride ion transfer.

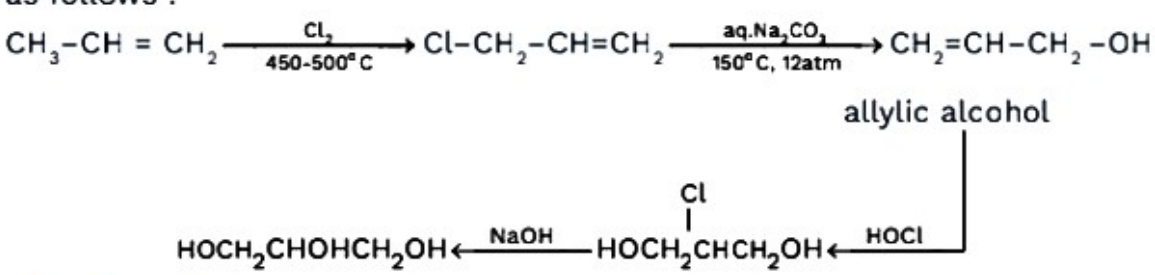
primary alcohol corresponding to the acid part of the ester.



10. Hydroxylation of alkenes :



11. The only important trihydric alcohol is glycerol (propane-1,2,3-triol) which is prepared as follows :



Chemical Properties :

1. Reaction with halogen acids :

R – OH + HX RX + H_2O : (R-group may rearrange)

Reactivity of HX : HI > HBr > HCl
Reactivity of ROH : Allyl > benzyl > 3° > 2° > 1°

Example :

(i) $CH_3CH_2CH_2CH_2-OH \xrightarrow[\Delta]{HCl,ZnCl_2} CH_3CH_2CH_2CH_2-Cl$

(ii) $H_3C-\underset{\underset{OH}{|}}{CH}-CH_3 \xrightarrow[\text{reflux}]{NaBr \text{ or } H_2SO_4} H_3C-\underset{\underset{Br}{|}}{CH}-CH_3$

(iii) $H_3C-\overset{\overset{CH_3}{|}}{\underset{\underset{OH}{|}}{C}}-CH_3 \xrightarrow{Conc.HCl} H_3C-\overset{\overset{CH_3}{|}}{\underset{\underset{Cl}{|}}{C}}-CH_3$

2. Reaction with phosphorus trihalides :

$3\ R{-}OH + PX_3 \rightarrow 3\ R{-}X + H_3PO_3$ ($PX_3 = PBr_3, PI_3$)

Example :

(i) 

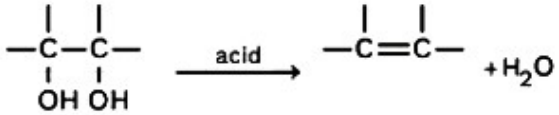
1-Phenyl ethanol 1-Bromo-1-Phenyl ethane

(ii) 

3 – Methyl – 1 – butanol 3 – Methyl – 1 – bromo butane

3. Dehydration :

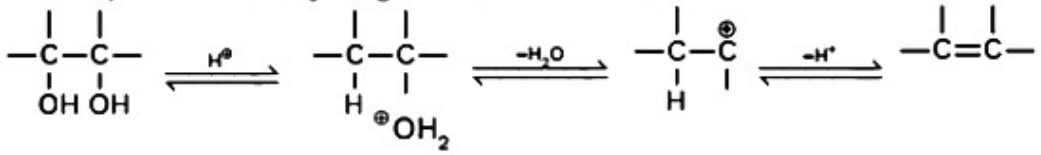
Alcohols may be dehydrated to alkenes, the ease of dehydration being 3° > 2° > 1°. Dehydration can be done by heat alone (400-800°C), but in the presence of a catalyst, lower temperature is required e.g., when passed over heated alumina, t-alcohols are dehydrated at about 150°C, s-alcohols at 250°C and p-alcohols at 350°C. Also p-alcohols are dehydrated by conc. H_2SO_4 at about 170°C and secondary and tertiary alcohols by boiling dilute sulphuric acid.



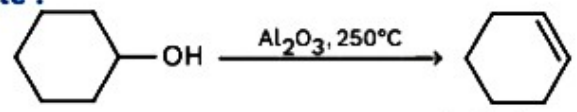
Reactivity of ROH : 3° > 2° > 1°.

Mechanism :

According to the commonly accepted mechanism first step is the formation of the protonated alcohols, $R{-}\overset{\oplus}{O}H_2$ followed by it's slow dissociation into a carboniumion. The last step in the fast expulsion of a hydrogen ion from the carbonium ion to form an alkene.



Example :

(i) 

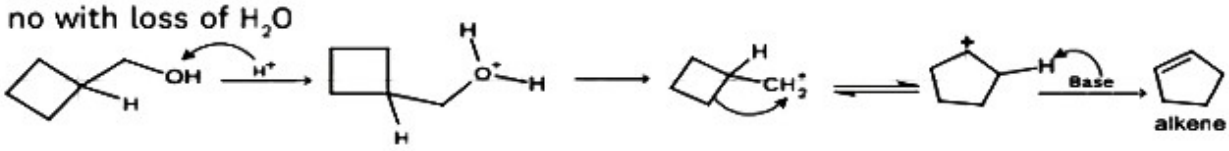
Cyclo hexanol Cyclo hexene

(ii) $CH_3-CH_2CH_2CH_2OH \xrightarrow[\Delta]{H_2SO_4} CH_3CH = CHCH_3 + CH_3CH_2CH = CH_2$

Butanol — 2-Butene — 1-Butene (Major)

Q71 **Give the product and write a mechanism for the acid dehydration of cyclobutycarbinol.**

A71 Expansion of a four – to a five-membered ring which relieves ring strain is synchrono with loss of H_2O



4. Reaction as Acids :

When reacts with active metals alcohols act as acids.

$R - OH + M \longrightarrow \overset{\ominus}{R}O\overset{\oplus}{M} + \frac{1}{2} H_2$ M = Na, K, Mg, Al etc.

Reactivity of ROH : $CH_3OH > 1^\circ > 2^\circ > 3^\circ$

Example :

$CH_3CH_2OH \xrightarrow{Na} CH_3CH_2ONa + \frac{1}{2} H_2$

Sodium ethoxide

$H_3C-\underset{\underset{OH}{|}}{\overset{\overset{CH_3}{|}}{C}}-H \xrightarrow{Al} (Me_2CHO)_3 Al$

Al, isopropoxide

$Me_3COH \xrightarrow{K} Me_3CO^-K^+$

pot. tert-butoxide

5. Ester formation :

Alcohols react with organic and inorganic acids to form ester

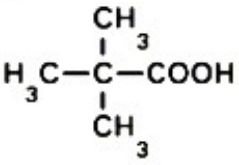
$RCOOH + R'-OH \longrightarrow R-COO-R' + H_2O$

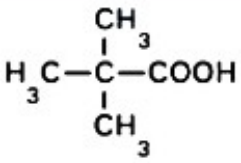
The mechanism is in general bimolecular and if it so in all cases, the order of reactivity of an alcohol with a given acid is primary alcohol > secondary > tertiary

6. Primary and secondary alcohols may be acetylated with acetyl chloride e.g. ethanol gives ethyl acetate.

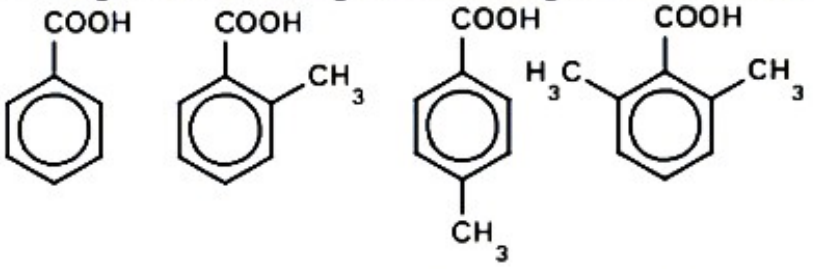
$H_3C-\overset{\overset{O}{\|}}{C}-Cl + C_2H_5OH \longrightarrow H_3C-\overset{\overset{O}{\|}}{C}-OEt + HCl$

Q72 **Arrange the following in increasing order of esterification :**

$HCOOH,\ CH_3COOH,\ CH_3CH_2COOH,$ 

A72 $HCOOH > CH_3COOH > CH_3CH_2COOH >$ 

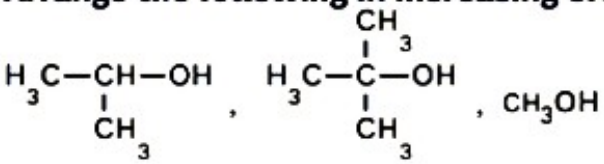
Q73 **Arrange the following in increasing order of esterification :**



(i) (ii) (iii) (iv)

A73 IV < II < III < I

Q74 **Arrange the following in increasing order of esterification**



(i) (ii) (iii)

A74 III > I > II $\left[\text{Size of alcohol } \alpha \frac{1}{\text{Rate of reaction}}\right]$

7. Oxidation of alcohols :

Alcohols may be oxidised, and the products of oxidation depend on the class of the alcohol and on the nature of the oxidising agent.

(i) A primary alcohol on oxidation first gives an aldehyde, and this, on further oxidation, gives an acid. Both aldehyde and acid contain the same number of carbon atoms as the original alcohol.

$$R{-}CH_2{-}OH \xrightarrow{[O]} R\,CHO \xrightarrow{[O]} RCOOH$$

(ii) A secondary alcohol, on oxidation, first gives a ketone with the same number of carbon atoms as the original alcohol ketones are fairly difficult to oxidise, but prolonged action of the oxidising agents produces a mixture of acids, each containing fewer carbon atoms than the original alcohol.

$$CH_3CH_2CH_2{-}\underset{\displaystyle OH}{\underset{|}{CH}}{-}CH_3 \xrightarrow{[O]} CH_3{-}CH_2{-}CH_2{-}\overset{\displaystyle O}{\overset{\|}{C}}{-}CH_3 \xrightarrow{[O]} CH_3CO_2H + CH_3CH_2CO_2H$$

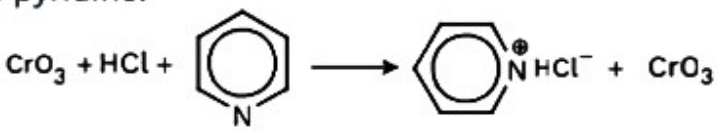
Tertiary alcohols are most difficult to oxidise among the three classes of alcohols.

The oxidising agents generally used for oxidising alcohols are : acid dichromate, acid or alkaline potassium permanganate, dilute nitric acid, aqueous $K_2Cr_2O_7$, CrO_3 in general acetic acid, CrO_3 in pyridine.

A very useful Cr(VI) reagent in pyridinium chlorochromate (PCC), which is prepared by dissolving CrO_3 in hydrochloric acid and by subsequent treatment of the solution with pyridine.

Points to remember

With no hydrogen attached to the carbonyl carbon, tertiary alcohols are not oxidized at all under alkaline conditions. If acid is present, they are rapidly dehydrated to alkenes, which are then oxidised.



PCC oxidised	1°–alcohols → Aldehydes
	2°–alcohols → Ketones
Allylic 1°–alcohols →	Corresponding aldehydes.

8. When the vapours of the alcohol are passed over copper at 300°C, the three classes of alcohols differ in their behaviour.

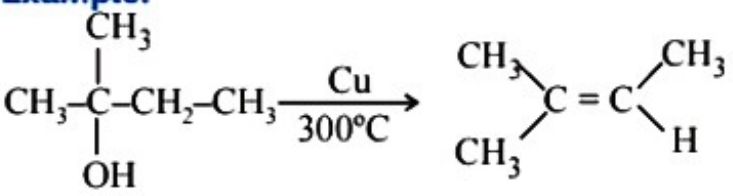
(i) $CH_3CH_2OH \xrightarrow[300°C]{Cu} CH_3CHO + H_2$, here the 1°-alcohol is dehydrogenated to an aldehyde.

(ii) A secondary alcohol is dehydrogenated to a ketone, e.g.

$$CH_3{-}\underset{\displaystyle OH}{\underset{|}{CH}}{-}CH_3 \xrightarrow[300°C]{Cu} CH_3{-}\overset{\displaystyle O}{\overset{\|}{C}}{-}CH_3 + H_2$$

(iii) But the tertiary alcohol is dehydrated to an alkene,

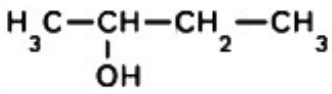
Example:

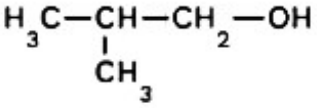


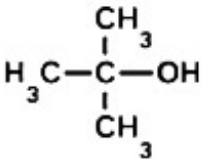
Q75 **Draw the structures of all the isomeric alcohols having the molecular formula $C_4H_{10}O$**

A75

(i) $CH_3CH_2CH_2CH_2OH$ — n-Butyl alcohol

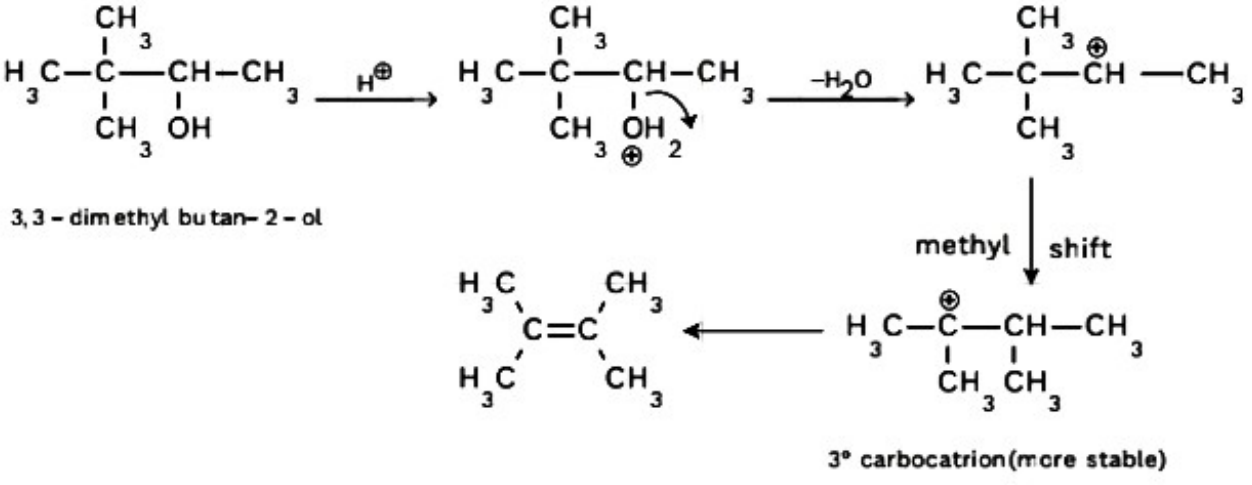
(ii)  — see-Butyl alcohol

(iii)  — Isobutyl alcohol

(iv)  — tert-Butyl alcohol

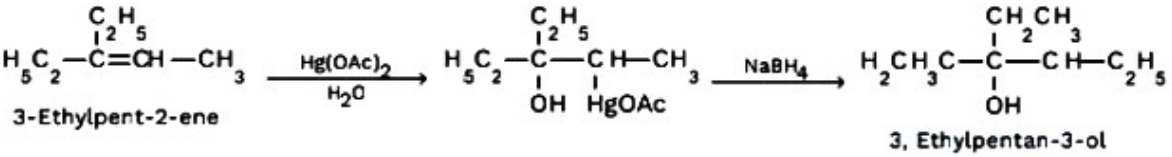
Q76 **3,3-dimethyl butan-2-ol loses a molecule of water in the presence of concentrated H_2SO_4 to give tetra methyl ethylene as the major product. Suggest a suitable mechanism.**

A76



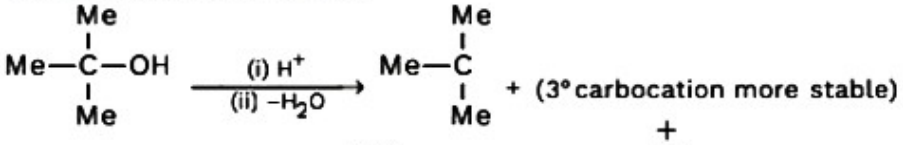
Q77 **Give the structures of the major organic product when 3-ethylpent-2-ene is reacted with $Hg(OAc)_2/H_2O$, $NaBH_4$.**

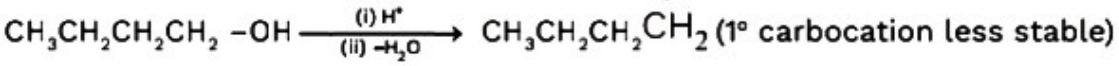
A77



Q78 **Acid catalysed dehydration of t-Butanol is faster than that of n-butanol. Give reasons in one or two sentences.**

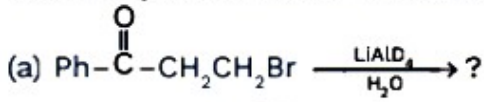
A78 Acid catalysed dehydration of alcohols occurs through the formation of carbocation as an intermediate.

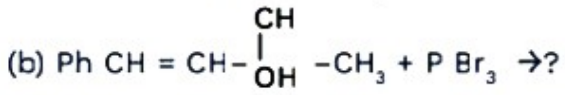




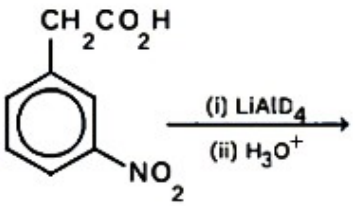
Since t-Butyl carbocation is more stable than n-Butyl carbocation, therefore acidcatalysed dehydration of t-Butanol occurs faster than n-butanol.

Q79 **Give the product of each reaction.**

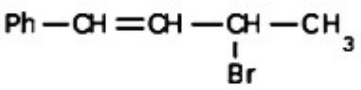
(a) 

(b) 

(c) 

(d) 

A79 (a) $PhCD(OH)CH_2CH_2D$

(b) 

(c) $CH_3CH_2CH = CH\text{-}CHO$

(d) 